



DC – AI – Routing – Switching – Security – Wireless – Automation

Guide to the EVPN VXLAN-based OISM on PTX

An HPE Juniper Networking Techpost

Ramdas Machat

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Contents

1	EVPN-VXLAN OISM on PTX Routers	3
1.1	Introduction	3
1.2	Document Scope	3
1.2.1	Multicast in EVPN and the problem with the CRB Model	3
1.2.2	Optimized Inter-Subnet Multicast (OISM)	4
1.2.3	Optimized Inter-subnet Multicast within Fabric	5
1.2.4	OISM – Inter-Subnet Multicast from outside fabric (Classic L3)	6
1.3	Topology and Testing	7
1.3.1	Underlay, Overlay and Customer Equipment (CE) Configuration	10
1.4	OISM Specific and Global Configuration	10
1.4.1	Originate SMET-Route-on-RevBD Configuration	10
1.4.2	Multicast snooping to install L2 (*,G) forwarding routes in PFE	10
1.4.3	OISM Global Configuration	10
1.4.4	OISM – PIM-GW as Classic L3 Interface	10
1.4.5	OISM PIM-GW with M-VLAN	35
1.5	Operational Overview	38
1.5.1	OISM States on BL & PIM-GW	39
1.6	Configuration Snippets	45
1.6.1	Access / Underlay Configuration	45
1.6.2	Overlay Configuration	51
1.6.3	CE Configuration - SP/EP style	53
1.7	Glossary	55
1.8	Acknowledgments	55

List of Tables

6	Multicast Flow details	11
7	CLI commands to verify OISM status on each node	17

1 EVPN-VXLAN OISM on PTX Routers

Ramdas Machat - 09/04/2023

Guide to the EVPN VXLAN based Optimised Inter-subnet Multicast (OISM) on Express4 based PTX10k platforms.

This document is co-written by Ramdas Machat and Abdul Nasir M.

1.1 Introduction

Handling unicast traffic is easy within the EVPN-VXLAN Data Centre Fabric (DCF) with no worries of handling the BUM, Inter-subnet routing and duplicate traffic. For multicast, that is not always the case. There are challenges in handling multicast in the Centrally Routed Bridge model with a single home or multi-homed environment. Over the time, multiple technologies emerged to overcome these challenges and provide the most efficient way to deliver multicast traffic within and outside of the DCF. This document touches upon the Optimized Inter-Subnet Multicast (OISM) and details the configurations and operational procedures to deploy Express4 based PTX10k in the DCF.

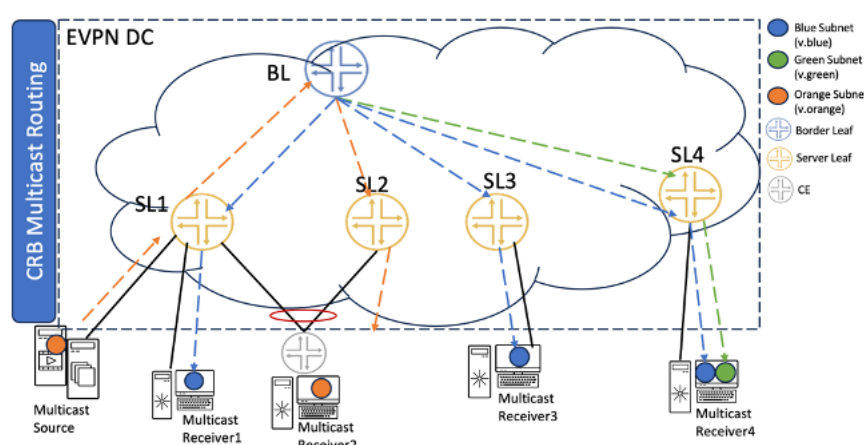
1.2 Document Scope

This document assumes that the reader understands EVPN-based Selective Multicast Ethernet Tag (SMET) and Inclusive Multicast Ethernet Tag (IMET) traffic forwarding in a data centre (DC) without optimization. We will test EVPN OISM on Express4 based PTX platforms covering “BD everywhere” model with configuration and operational aspects. “OISM with BD not everywhere” is not covered in this document.

1.2.1 Multicast in EVPN and the problem with the CRB Model

EVPN-VXLAN helps stretching L2 networks across sites at scale and is widely deployed in enterprise data centre networks. Optimized multicast traffic delivery has been supported and deployed in a Centrally Routed Bridge (CRB) Model for a while now.

SMET and Assisted Replication (AR) optimizes multicast traffic within a subnet (VLAN) in the CRB model, inter-subnet multicast traffic forwarding still could be handled better in terms of avoiding traffic hair-pinning and reducing core-bandwidth utilization. This is particularly required when there are many subnets with listeners for a multicast flow.



In Figure 1, the traffic from Server Leaf 1 (SL-1) to “Multicast receiver1” goes through the Border Leaf (BL) instead of routing it locally, which consumes the core bandwidth based on the number of flows. This poses a replication load on BL and increases the number of copies of the packet being sent on the core. This is called traffic hair-pinning and it is inefficient: it increases replication on BL and adds latency. If SL1 could route the traffic locally, it would reduce end-to-end latency

For details on the inter-subnet routing forwarding rules refer to the [DayOne OISM EVPN book](#).

1.2.2 Optimized Inter-Subnet Multicast (OISM)

Optimized Inter-Subnet Multicast function introduces the following traffic forwarding benefits:

- In OISM model the traffic is always switched to the remote receivers using the source VLAN and the downstream multicast receiver gets routed traffic. For example, if multicast traffic is sourced at SL1 (v.orange) and the receiver is on SL3 (v.blue), then the traffic will be EVPN-switched towards SL3 using v.orange only and then routed from v.orange to v.blue locally at SL3.
- It reduces latency.
- OISM makes the network more reliable as it creates a backup path for traffic in case one Border Leaf fails.
- Multicast traffic received from outside fabric is delivered within fabric on Supplementary Bridge Domain (SBD) to receiver’s leaves which further route multicast traffic to directly connected receivers.

Table 1 contains the list of supported features for the PTX platforms in 22.3R1 EVO. Please refer to OISM Documentation for the latest and more details updates.OISM Feature Support:

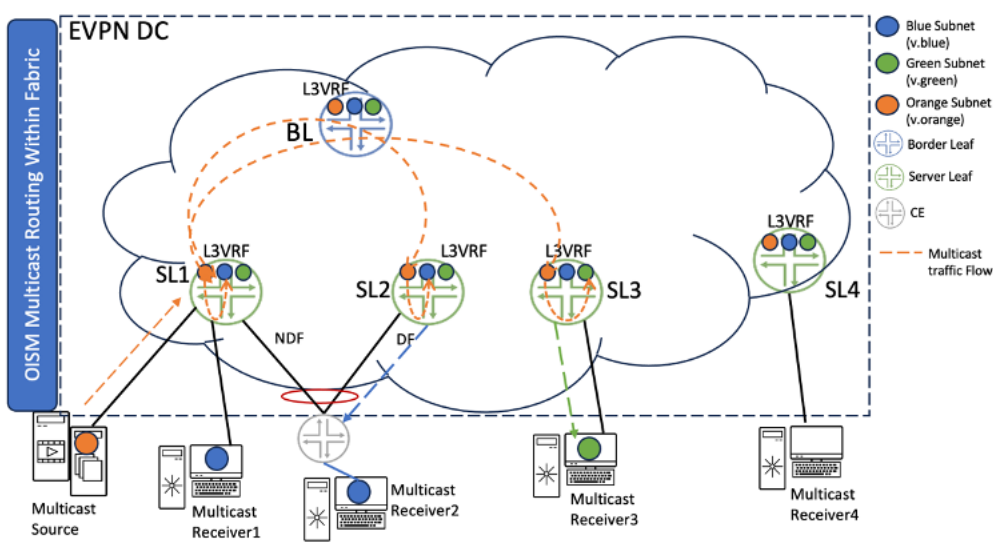
Features	Supported on PTX10000 Chassis and PTX10001-36MR
Intra-VLAN Multicast traffic with IGMP Snooping with Multi-homing consideration for EVPN over VXLAN	YES
Inter-VLAN Multicast traffic and IRB for EVPN over VXLAN	YES
OISM support for EVPN over VXLAN with BDs everywhere for IGMP v2/v3	YES
Role of Server Leaf	YES
Role of Border Leaf with PIM-GW over EVPN-VXLAN IRB/L3 interface/non-EVPN IRB	YES
AE/non-AE/ECMP interface will be supported as underlay/OIF in the multicast route	YES
EVPN Type6 SMET Route to selectively send or receive traffic based on the presence of active receivers	YES

Features	Supported on PTX10000 Chassis and PTX10001-36MR
EVPN A/A Multi-homing by using BGP EVPN Type 7/8 NLRI to sync IGMP joins/leaves received on multi-homed interfaces between multi-homing peer PEs	YES
Distributed DR for EVPN over VXLAN Multicast traffic with OISM	YES
SP and EP style configurations	YES
EVPNomPLS Snooping	NO
Assisted Replication for both Leaf and Replicator role	NO
EVPNomVXLAN with ERB Multicast	NO
OISM with BDs not everywhere	NO
MLD Snooping v1/v2	NO
IGMPv2 and IGMPv3 co-existence	NO
IGMPv2 and IGMPv3 Revenue BDs in same tenant VRF	NO

1.2.3 Optimized Inter-subnet Multicast within Fabric

In Figure 2 below, “Multicast Source” is behind SL1 on v.orange and the multicast traffic is originated from there. The dotted arrow lines in the figure indicate the multicast traffic flow. The receivers for this multicast traffic are:

1. Single-homed SL1 on v.blue VLAN [*Multicast receiver 1*]
2. Multi-homed SL1/SL2 on v.blue VLAN [*Multicast receiver 2*]
3. Single-homed SL3 on v.blue VLAN [*Multicast receiver 3*]



- First SL1 will route the traffic locally from IRB of v.orange VLAN to IRB of v.blue VLAN and send it towards “*Multi-homed Receiver 1*”

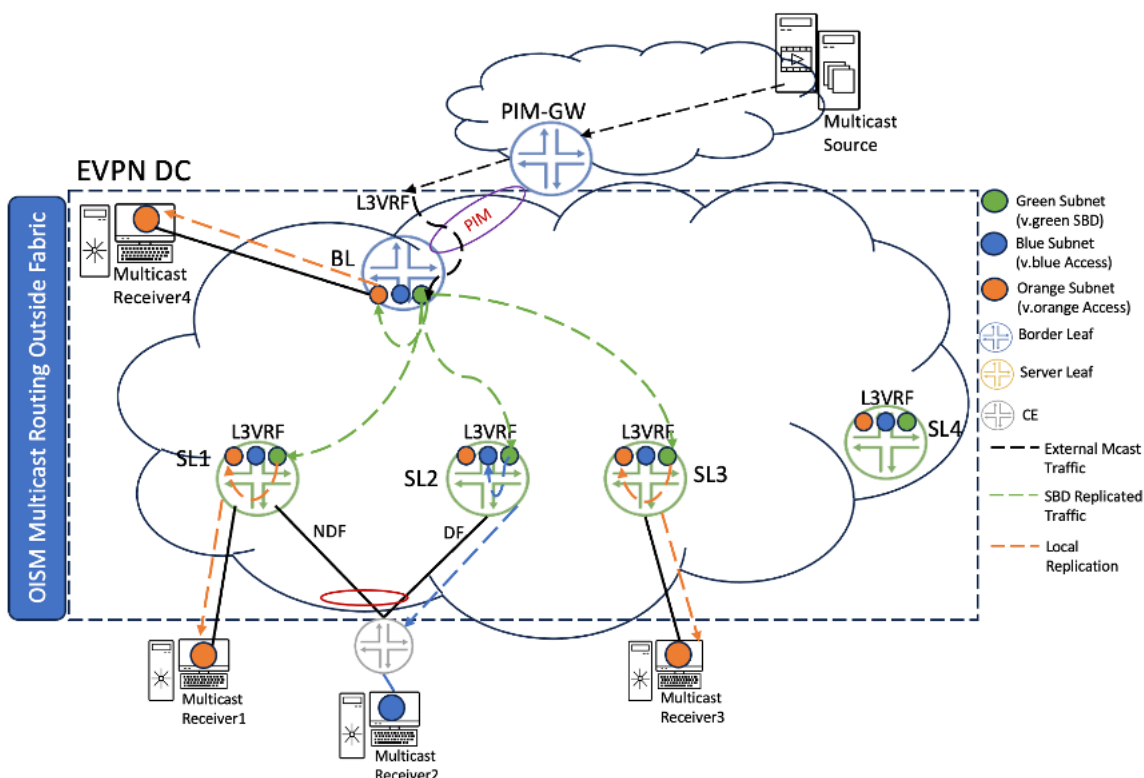
- One copy of multicast traffic will be EVPN switched towards SL2 on source VLAN [v.orange] and SL2 will send traffic towards “*Multi-homed Receiver 2*” as SL2 is the EVPN DF.
- One copy of traffic will be EVPN switched towards SL3 on source VLAN [v.orange] and SL3 would route it to IRB of v.green VLAN locally and sent towards “*Multicast Receiver 3*”

Please note: SL2 and SL3 will not send traffic back to the core post routing traffic from source VLAN to receiver VLAN.

1.2.4 OISM – Inter-Subnet Multicast from outside fabric (Classic L3)

In the scenario of Figure 3 Border Leaf node is acting as PIM External Gateway (PEG). The dotted arrow lines in the figure represent the multicast traffic flow. Here the multicast source is external to DC fabric and the receivers are:

1. Single-homed SL1 on v.orange VLAN [*Multicast receiver 1*]
2. Multi-homed SL1/SL2 on v.blue VLAN [**Multicast receiver 2*]*
3. Single-homed SL3 on v.orange VLAN [*Multicast receiver 3*]
4. Single-homed BL on v.orange VLAN [*Multicast receiver 4*]



Traffic flow from the External world to the EVPN DC is described below.

- Multicast Traffic pulled from an external source will land on BL (by virtue of pim join sent towards the external source from BL)
- Since there is a local receiver “*Multicast receiver 4*” on BL, it will locally route the traffic to IRB of v.orange VLAN and send it towards “*Multicast receiver 4*”.
- BL will also route the traffic to IRB of Supplementary Bridge Domain (SBD) v.green VLAN, since it has received Type6 SMET signalling on SBD VLAN.

- Post routing to IRB of SBD VLAN, one copy of multicast traffic will be EVPN switched towards SL1, SL2, and SL3 on v.green VLAN.
- On SL1, traffic will be locally routed from IRB of SBD v.green VLAN to IRB of orange VLAN and sent towards “*Multicast Receiver 1*”.
- Both SL1 and SL2 will route the traffic from IRB of v.green VLAN to IRB of v.orange VLAN, but only EVPN DF among SL1 or SL2 will send traffic towards “*Multicast Receiver 2*”. In this case, SL2 is the EVPN DF.
- On SL3, traffic will be locally routed from IRB of SBD v.green VLAN to IRB of v.orange VLAN and sent towards “*Multicast receiver 3*”.

Please note: SL1, SL2, SL3, and BL will not send traffic back to the core post routing traffic from source VLAN to receiver VLAN. It is not recommended to host any server (unicast/multicast) on SBD to avoid any anomalies like traffic duplication.

1.3 Topology and Testing

We have validated EVPN OISM service on PTX10K platforms powered by Express 4 ASIC and Junos EVO on 400GE interfaces.

Figure 4 illustrates the reference topology below, used to test OISM with multicast sources and receivers within the data center and outside of the data center domain with PIM-GW as Classic L3 or MVLAN.

PTX platform has been validated in three different roles (see Table 2).

OISM Nodes and Roles:

Node	Details
Border Leaf	PTX10K can be placed as Border Leaf to receive multicast traffic from outside world or to send multicast traffic to outside world from the source within fabric
Leaf Spine	Spine devices are underlay of EVPN Fabric as IP transit devices. The leaf spines might also act as Route Reflectors (RR) in the fabric. PTX10k is capable of playing this role
Server Leaf	PTX10K can be placed as Server Leaf to perform local multicast routing. In this role, the multicast traffic from other SL devices received on source VLAN is routed to receiver VLAN. The multicast traffic from external world (through BL devices) is received on SBD VLAN and is routed to receiver VLAN on the SLs.

For external Multicast we have validated three scenarios (see Table 3).

Mode	Details
External IRB – MVLAN	The PIM-GW node will be multihomed to the BL nodes with EVPN-VXLAN. So, the PIM-GW will have BGP peering with the BL nodes.
Internal Classic L3 Interface	On the PIM-GW the Multicast source/receivers are connected using a regular L3 interface. The connectivity between PIM-GW and BL is a regular L3 Interface with PIM running.
L3 Over IRB	On the PIM-GW the Multicast source/receivers are connected using L2, terminated on the IRB which is running PIM protocols.

For delivering multicast within the fabric, we have tested scenarios outlined in Table 4.

OISM Terminology:

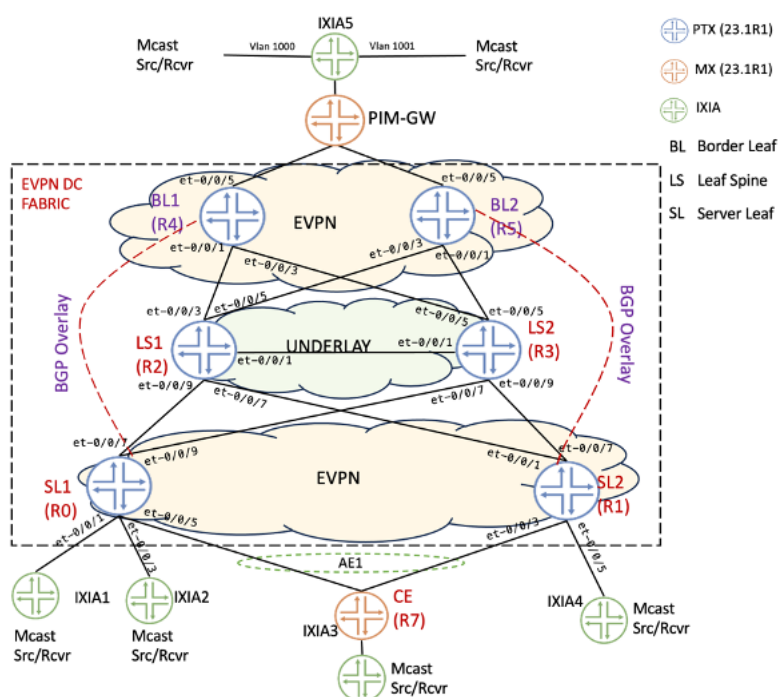
Bridge Domain / VLAN	Description	Configured on
Multicast VLAN(M-VLAN)	A VLAN in the EVPN fabric with associated IRB interfaces that connect the fabric to an external multicast router. This VLAN and IRB interface enable traffic flow between devices inside and outside the fabric.	Border leaf devices
Extra non-EVPN VLAN	(Non-EVPN IRB method for external multicast). An extra VLAN that isn't in the EVPN instances in the fabric. You configure associated IRB interfaces in the tenant L3 VRF instances. This VLAN and IRB interface enable multicast traffic flow between devices inside the fabric and devices outside the fabric.	Border leaf devices
Revenue Bridge Domains(VLANs)	Bridge domains for subscribers to the services that the fabric provides. You configure the revenue bridge domains as VLANs in the fabric.	All OISM devices

Bridge Domain / VLAN	Description	Configured on
Supplemental Bridge Domain(SBD)	Bridge domain that enables support for external multicast traffic and implements SMET optimization in the EVPN core.	All OISM devices

Table 5 summarises information about router models, roles and Junos releases, helper MX platforms and traffic generators used in the test bed.

Topology devices details:

Device Name	Device Model	Software Release
R0-Server Leaf1	PTX10001-36MR	23.1R1.8-EVO
R1-Server Leaf2	PTX10001-36MR	23.1R1.8-EVO
R2-Spine Leaf1	PTX10001-36MR	23.1R1.8-EVO
R3-Spine Leaf2	PTX10001-36MR	23.1R1.8-EVO
R4-Border Leaf1	PTX10001-36MR	23.1R1.8-EVO
R5-Border Leaf1	PTX10001-36MR	23.1R1.8-EVO
R6-External	MX480	22.4R1.9
R7-CE1	MX480	22.4R1.9
R8, R9, R10, R11 – Multicast Source/Receivers	Ixia	8.20



1.3.1 Underlay, Overlay and Customer Equipment (CE) Configuration

All the devices including SL, BL, LS and PIM-GW are configured with OSPF as an IGP for underlay layer. For EVPN-VXLAN overlay and exchanging EVPN routes SL and BL nodes are establishing full meshed MS-iBGP sessions for evpn signalling. The Sample configuration of the underlay protocols on PTX and CE configuration of MX router (including SP/EP style configuration) is covered as part of Access/Underlay Configuration & CE configuration - SP/EP Style (check the config snippets at the end of this article).

1.4 OISM Specific and Global Configuration

1.4.1 Originate SMET-Route-on-RevBD Configuration

On BL and SL devices, configure the following knob to originate Type-6 on RevBD where the local IGMP report is received. Without this knob, Type-6 is originated only on SBD alone, as per [draft-ietf-bess-evpn-irb-mcast](#). This knob is provided to enable interop with other vendor devices.

```
"set routing EVPN_OISM protocols evpn oism originate-smet-on-revenue-vlan-too"
```

Without this knob, Type-6 is originated on SBD only. This knob enables interoperability with other vendor devices that do not support OISM.

1.4.2 Multicast snooping to install L2 (*,G) forwarding routes in PFE

By default for OISM, the L2(*,G) forwarding routes will not be installed in the PFE. To override the default behaviour, use the following knob. This is for scaling considerations for OISM. Hence, use this knob to install L2(*,G) routes to PFE

```
set routing-instance <instance-name> multicast-snooping-options oism install-star-g-routes"
```

This improves the scaling in OISM deployments.

1.4.3 OISM Global Configuration

Tunnel-termination is not enabled by default on PTX platforms. User has the option to enable tunnel-termination either globally or at the underlay interface level for VXLAN tunnel termination to work. Use the below command:

```
"set forwarding-options tunnel-termination"
```

Enable OISM “BD everywhere” mode on PTX SL and BL nodes by issuing the following knob:

```
"set forwarding-options multicast-replication evpn irb oism"
```

1.4.4 OISM – PIM-GW as Classic L3 Interface

Server Leaves (R0, R1) in Figure 4 have directly connected multicast sources and receivers. CE (R7) is multi-homed to SL1 (R0) and SL2 (R1) as active/active ESI LAG with directly connected multicast source/receivers. Border Leaf's (R4, R5) are connected to the PIM-GW (R6) with multicast sources, receivers attached.

Configuration Overview

Each node configuration is detailed towards the end of this document in the Configuration Snippet section with name configured outlined as Access/Underlay, Overlay. Follow below steps to bring up the setup for OISM testing:

1. Bring up the underlay on the respective SL, BL, PIM-GW using the Access/Underlay section that includes the provisioning of Interfaces (Physical & Loopback), Bundles & IGP. Make sure everything works end-to-end
2. Once the underlay is all up and reachability is established. Make the overlay configurations
3. Use the section below to bring up the OISM related configuration on SLs, BLs. Use the given configuration under each sections “SL Configuration” and “PEG Configuration” respectively.
4. Once we have all the configurations done, simulate the IGMP hosts from IXIA for multicast joins and to send multicast traffic.
5. There are multicast sources and receivers connected to PIM-GW, SL1, SL2 and multihomed CE. There are two service VLANs in the DC VLAN100, VLAN101 and one SBD VLAN900 in the topology. There are two multicast group 225.1.1.10, 225.1.1.20 and receivers are associated in VLAN100, VLAN101, VLAN1000, VLAN10001. Below Table 5 lists down the various flows generated for this test.

Table 6: Multicast Flow details

Multicast Flow	Source	Multicast Group	Receivers (VLAN & Node)	Traffic Rate
SL1-VLAN-101	VLAN101 (101.1.1.3)	225.1.1.10	SL1(100),SL2(100),PIM1000pps GW(1000),MHCE(100)	
PIM-GW-VLAN-1000	VLAN1000 (25.1.1.2)	225.1.1.10	SL1(100),SL2(100),PIM1000pps GW(1000),MHCE(100)	
PIM-GW-VLAN-1001	VLAN1001 (25.1.2.2)	225.1.1.10	SL1(100),SL2(100),PIM1000pps GW(1000),MHCE(100)	
SL1-VLAN-100	VLAN100 (100.1.1.3)	225.1.1.20	SL1(100,101),SL2(100),PIM1000pps GW(1000),MHCE(100,101)	

SL Configuration

Below configuration is focused on the OISM specific configuration on the SLs and the common configurations including underlay/overlay are in the configuration snippets.

Table: Multicast Flow details

SL1 (R0)

```

1 R0-SL1# show routing-instances
2 EVPN_OISM {
3     instance-type mac-vrf;
4     protocols {
5         igmp-snooping {
6             vlan v100 {
7                 proxy;
8             }
9             vlan v101 {
10                proxy;

```

```

11     }
12     vlan v900 {
13         proxy;
14     }
15 }
16 evpn {
17     encapsulation vxlan;
18 }
19 }
20 vtep-source-interface lo0.0;
21 service-type vlan-aware;
22 route-distinguisher 1.1.1.1:1;
23 vrf-target target:1:1;
24 multicast-snooping-options {
25     oism {
26         install-star-g-routes;
27     }
28 }
29 vlans {
30     v100 {
31         vlan-id 100;
32         interface ae1.100;
33         interface et-0/0/1.100;
34         interface et-0/0/3.100;
35         l3-interface irb.100;
36         vxlan {
37             vni 100;
38         }
39     }
40     v101 {
41         vlan-id 101;
42         interface et-0/0/1.101;
43         interface ae1.101;
44         interface et-0/0/3.101;
45         l3-interface irb.101;
46         vxlan {
47             vni 101;
48         }
49     }
50     v900 {
51         vlan-id 900;
52         l3-interface irb.900;
53         vxlan {
54             vni 900;
55         }
56     }
57 }
58 }
59 OISM_VRF {
60     instance-type vrf;
61     protocols {
62         evpn {
63             oism {
64                 supplemental-bridge-domain-irb irb.900;
65             }
66         }
67         pim {
68             passive;
69             interface all;
70             interface irb.900 {
71                 accept-remote-source;
72             }
73         }
74     }
75     interface irb.100;
76     interface irb.101;
77     interface irb.900;
78     interface lo0.1;
79     route-distinguisher 1.1.1.1:2;
80     vrf-target target:1:2;
81 }
82
83 R0-SL1# show forwarding-options
84 tunnel-termination;
85 multicast-replication {
86     evpn {
87         irb oism;
88     }
89 }

```

SL2 (R1)

```

1 R1-SL2# show routing-instances
2 EVPN_OISM {
3     instance-type mac-vrf;
4     protocols {
5         igmp-snooping {
6             vlan v100 {
7                 proxy;
8             }
9             vlan v101 {
10                proxy;
11            }
12            vlan v900 {
13                proxy;
14            }
15        }
16        evpn {
17            encapsulation vxlan;
18        }
19    }
20    vtep-source-interface lo0.0;
21    service-type vlan-aware;
22    route-distinguisher 1.1.1.2:1;
23    vrf-target target:1:1;
24    multicast-snooping-options {
25        oism {
26            install-star-g-routes;
27        }
28    }
29    vlans {
30        v100 {
31            vlan-id 100;
32            interface ae1.100;
33            interface et-0/0/5.100;
34            l3-interface irb.100;
35            vxlan {
36                vni 100;
37            }
38        }
39        v101 {
40            vlan-id 101;
41            interface ae1.101;
42            interface et-0/0/5.101;
43            l3-interface irb.101;
44            vxlan {
45                vni 101;
46            }
47        }
48        v900 {
49            vlan-id 900;
50            interface et-0/0/5.900;
51            l3-interface irb.900;
52            vxlan {
53                vni 900;
54            }
55        }
56    }
57 }
58 OISM_VRF {
59     instance-type vrf;
60     protocols {
61         evpn {
62             oism {
63                 supplemental-bridge-domain-irb irb.900;
64             }
65         }
66         pim {
67             passive;
68             interface all;
69             interface irb.900 {
70                 accept-remote-source;
71             }
72         }
73     }
74     interface irb.100;
75     interface irb.101;
76     interface irb.900;
77     interface lo0.1;
78     route-distinguisher 1.1.1.2:2;

```

```

79     vrf-target target:1:2;
80 }
81
82 R1-SL2# show forwarding-options
83 tunnel-termination;
84 multicast-replication {
85     evpn {
86         irb oism;
87     }
88 }

```

PEG Configuration/ SBD Configuration

The BL Devices in the OISM setup are made EVPN PEG using configuration “protocols evpn oism pim-evpn-gateway” to pull the traffic from the SLs so that BLs can route the traffic to external receivers. SL devices are not required to be configured as PEG.

To configure SBD on all OISM PEs use this configuration “supplemental-bridge-domain-irb irb.” Here It can be configured at the L3 VRF level specifying the IRB associated with the SBD. The same VNI must be configured for the SBD on all the PEs for a particular L3-VRF. The SBD IRB interface should be configured in this L3-VRF instance.

BL1 (R4)

```

1  R4-BL1> show configuration routing-instances
2  EVPN_OISM {
3      instance-type mac-vrf;
4      protocols {
5          igmp-snooping {
6              vlan v100;
7              vlan v101;
8          }
9          evpn {
10             encapsulation vxlan;
11             default-gateway no-gateway-community;
12         }
13     }
14     vtep-source-interface lo0.0;
15     service-type vlan-aware;
16     route-distinguisher 1.1.1.5:1;
17     vrf-target target:1:1;
18     multicast-snooping-options {
19         oism {
20             install-star-g-routes;
21         }
22     }
23     vlans {
24         v100 {
25             vlan-id 100;
26             l3-interface irb.100;
27             vxlan {
28                 vni 100;
29             }
30         }
31         v101 {
32             vlan-id 101;
33             l3-interface irb.101;
34             vxlan {
35                 vni 101;
36             }
37         }
38         v900 {
39             vlan-id 900;
40             l3-interface irb.900;
41             vxlan {
42                 vni 900;
43             }
44         }
45     }
46 }
47 OISM_VRF {
48     instance-type vrf;
49     protocols {
50         evpn {

```

```

51     oism {
52         supplemental-bridge-domain-irb irb.900;
53         pim-evpn-gateway;
54     }
55 }
56 ospf {
57     area 0.0.0.0 {
58         interface et-0/0/5.0;
59         interface lo0.1;
60         interface irb.100 {
61             passive;
62         }
63         interface irb.101 {
64             passive;
65         }
66         interface irb.900;
67     }
68 }
69 pim {
70     rp {
71         static {
72             address 1.1.1.7;
73         }
74     }
75     interface irb.100 {
76         distributed-dr;
77     }
78     interface irb.101 {
79         distributed-dr;
80     }
81     interface irb.900 {
82         accept-remote-source;
83     }
84     interface et-0/0/5.0;
85     interface lo0.1;
86 }
87 }
88 interface et-0/0/5.0;
89 interface irb.100;
90 interface irb.101;
91 interface irb.900;
92 interface lo0.1;
93 route-distinguisher 1.1.1.5:2;
94 vrf-target target:1:2;
95 }
96 R4-BL1> show configuration forwarding-options
97 tunnel-termination;
98 multicast-replication {
99     evpn {
100         irb oism;
101     }
102 }

```

BL2 (R5)

```

1 R5-BL2> show configuration routing-instances
2 EVPN_OISM {
3     instance-type mac-vrf;
4     protocols {
5         igmp-snooping {
6             vlan v100;
7             vlan v101;
8         }
9         evpn {
10             encapsulation vxlan;
11             default-gateway no-gateway-community;
12         }
13     }
14     vtep-source-interface lo0.0;
15     service-type vlan-aware;
16     route-distinguisher 1.1.1.6:1;
17     vrf-target target:1:1;
18     multicast-snooping-options {
19         oism {
20             install-star-g-routes;
21         }
22     }
23     vlans {
24         v100 {
25             vlan-id 100;

```

```

26         l3-interface irb.100;
27         vxlan {
28             vni 100;
29         }
30     }
31     v101 {
32         vlan-id 101;
33         l3-interface irb.101;
34         vxlan {
35             vni 101;
36         }
37     }
38     v900 {
39         vlan-id 900;
40         l3-interface irb.900;
41         vxlan {
42             vni 900;
43         }
44     }
45 }
46 }
47 OISM_VRF {
48     instance-type vrf;
49     protocols {
50         evpn {
51             oism {
52                 supplemental-bridge-domain-irb irb.900;
53                 pim-evpn-gateway;
54             }
55         }
56         ospf {
57             area 0.0.0.0 {
58                 interface et-0/0/5.0;
59                 interface lo0.1 {
60                     passive;
61                 }
62                 interface irb.100 {
63                     passive;
64                 }
65                 interface irb.101 {
66                     passive;
67                 }
68                 interface irb.900;
69             }
70         }
71         pim {
72             rp {
73                 static {
74                     address 1.1.1.7;
75                 }
76             }
77             interface irb.100 {
78                 distributed-dr;
79             }
80             interface irb.101 {
81                 distributed-dr;
82             }
83             interface irb.900 {
84                 accept-remote-source;
85             }
86             interface et-0/0/5.0;
87             interface lo0.1;
88         }
89     }
90     interface et-0/0/5.0;
91     interface irb.100;
92     interface irb.101;
93     interface irb.900;
94     interface lo0.1;
95     route-distinguisher 1.1.1.6:2;
96     vrf-target target:1:2;
97 }
98 R5-BL2> show configuration forwarding-options
99 tunnel-termination;
100 multicast-replication {
101     evpn {
102         irb oism;
103     }
104 }

```

Operational states and corresponding CLI commands Overview

The section below covers different verification CLI commands to check the status of the multicast forwarding on different nodes in the network including the SL, BL, PIM-GW. Table 6 shows the summary of the commands.

Table 7: CLI commands to verify OISM status on each node

Command	Description and Usage
show igmp snooping evpn database	Displays the entries related to bridge domain, vni, multicast group etc.
show igmp snooping evpn membership	Displays the multicast group entries along with the sources, OIFs and report.
show multicast route instance OISM_VRF extensive	This shows the multicast route entries for the active multicast traffic. Shows the downstream and upstream OIFs
show route table bgp.evpn.0 match-prefix 3*	This is check all Type 3 route generated by the router.
show route table bgp.evpn.0 match-prefix 6*	This is check all Type 6 routes generated and learnt by the router.
show route table bgp.evpn.0 match-prefix 7*	Type-7 is for Multicast/IGMP Join Sync
show route table bgp.evpn.0 match-prefix 8*	Type-8 is for Multicast/IGMP Leave
show evpn multicast-snooping next-hops	This is to find out the respective tunnels which is useful for debugging/troubleshooting
show evpn igmp-snooping database extensive l2-domain-id	Print the details on the IGMP snooping database specific to the VLAN, VNI etc.
show evpn igmp-snooping database extensive group	Print the details on Multicast details specific to the Multicast group.

OISM States on SL

The show cli commands below highlights the active receivers for the multicast group based on the sources on VLAN100, VLAN101 and the SBD 900.

SL1 (R0)

```

1 R0-SL1> show igmp snooping evpn database
2 Instance: EVPN_OISM
3   Bridge-Domain: v100, VN Identifier: 100
4     Group IP: 225.1.1.10, Source IP: 0.0.0.0, Access OIF Count: 2, Core NH: 47005
5   Bridge-Domain: v101, VN Identifier: 101
6     Group IP: 225.1.1.20, Source IP: 0.0.0.0, Access OIF Count: 2, Core NH: 47018
7   Bridge-Domain: v900, VN Identifier: 900
8
9 R0-SL1> show igmp snooping evpn membership
10 Instance: EVPN_OISM
11 Vlan: v100, EVPN-Core-NH: 47005
12 Learning-Domain: default
13 Interface: et-0/0/1.100, Groups: 0
14 Interface: ae1.100, Groups: 1
15   Group: 225.1.1.10

```

```

16      Group mode: Exclude
17      Source: 0.0.0.0
18      Type: Local
19 Interface: et-0/0/3.100, Groups: 1
20      Group: 225.1.1.10
21      Group mode: Exclude
22      Source: 0.0.0.0
23      Type: Local
24
25 Vlan: v101, EVPN-Core-NH: 47018
26 Learning-Domain: default
27 Interface: et-0/0/1.101, Groups: 0
28 Interface: ae1.101, Groups: 1
29      Group: 225.1.1.20
30      Group mode: Exclude
31      Source: 0.0.0.0
32      Type: Local
33 Interface: et-0/0/3.101, Groups: 1
34      Group: 225.1.1.20
35      Group mode: Exclude
36      Source: 0.0.0.0
37      Type: Local
38
39 Vlan: v900, EVPN-Core-NH: 47300
40
41 Learning-Domain: default
42
43 R0-SL1> show multicast route instance OISM_VRF extensive
44 Instance: OISM_VRF Family: INET
45
46 Group: 225.1.1.10
47   Source: 100.1.1.1/32
48   Upstream interface: irb.100
49   Downstream interface list:
50     irb.900
51   Number of outgoing interfaces: 1
52   Session description: Unknown
53   Statistics: 158 kbps, 323 pps, 572439 packets
54   Next-hop ID: 47682
55   Upstream protocol: Multicast
56   Route state: Active
57   Forwarding state: Forwarding
58   Cache lifetime/timeout: 360 seconds
59   Wrong incoming interface notifications: 1
60   Uptime: 00:09:51
61   Sensor ID: 0xf0000011
62
63 Group: 225.1.1.10
64   Source: 101.1.1.3/32
65   Upstream interface: irb.101
66   Downstream interface list:
67     irb.900 irb.100
68   Number of outgoing interfaces: 2
69   Session description: Unknown
70   Statistics: 157 kbps, 321 pps, 564135 packets
71   Next-hop ID: 47690
72   Upstream protocol: Multicast
73   Route state: Active
74   Forwarding state: Forwarding
75   Cache lifetime/timeout: 360 seconds
76   Wrong incoming interface notifications: 2
77   Uptime: 00:09:44
78   Sensor ID: 0xf0000013
79
80 Instance: OISM_VRF Family: INET6

```

SL2 (R1)

```

1 R1-SL2> show igmp snooping evpn database
2 Instance: EVPN_OISM
3   Bridge-Domain: v100, VN Identifier: 100
4     Group IP: 225.1.1.10, Source IP: 0.0.0.0, Access OIF Count: 2, Core NH: 47487
5   Bridge-Domain: v101, VN Identifier: 101
6     Group IP: 225.1.1.20, Source IP: 0.0.0.0, Access OIF Count: 2, Core NH: 47479
7   Bridge-Domain: v900, VN Identifier: 900
8
9 R1-SL2> show igmp snooping evpn membership
10 Instance: EVPN_OISM
11 Vlan: v100, EVPN-Core-NH: 47487
12 Learning-Domain: default

```

```

13 Interface: ae1.100, Groups: 1
14   Group: 225.1.1.10
15     Group mode: Exclude
16     Source: 0.0.0.0
17     Type: Remote
18 Interface: et-0/0/5.100, Groups: 1
19   Group: 225.1.1.10
20     Group mode: Exclude
21     Source: 0.0.0.0
22     Type: Local
23
24 Vlan: v101, EVPN-Core-NH: 47479
25 Learning-Domain: default
26 Interface: ae1.101, Groups: 1
27   Group: 225.1.1.20
28     Group mode: Exclude
29     Source: 0.0.0.0
30     Type: Remote
31 Interface: et-0/0/5.101, Groups: 1
32   Group: 225.1.1.20
33     Group mode: Exclude
34     Source: 0.0.0.0
35     Type: Local
36
37 Vlan: v900, EVPN-Core-NH: 47471
38
39 Learning-Domain: default
40 Interface: et-0/0/5.900, Groups: 0
41
42 R1-SL2> show multicast route instance OISM_VRF extensive
43 Instance: OISM_VRF Family: INET
44
45 Group: 225.1.1.10
46   Source: 100.1.1.1/32
47   Upstream interface: irb.100
48   Downstream interface list:
49     irb.900
50   Number of outgoing interfaces: 1
51   Session description: Unknown
52   Statistics: 477 kbps, 973 pps, 1682823 packets
53   Next-hop ID: 47709
54   Upstream protocol: Multicast
55   Route state: Active
56   Forwarding state: Forwarding
57   Cache lifetime/timeout: 360 seconds
58   Wrong incoming interface notifications: 0
59   Uptime: 00:28:09
60   Sensor ID: 0xf0000011
61
62 Group: 225.1.1.10
63   Source: 101.1.1.3/32
64   Upstream interface: irb.101
65   Downstream interface list:
66     irb.100 irb.900
67   Number of outgoing interfaces: 2
68   Session description: Unknown
69   Statistics: 474 kbps, 967 pps, 1677420 packets
70   Next-hop ID: 47719
71   Upstream protocol: Multicast
72   Route state: Active
73   Forwarding state: Forwarding
74   Cache lifetime/timeout: 360 seconds
75   Wrong incoming interface notifications: 2
76   Uptime: 00:28:02
77   Sensor ID: 0xf0000013
78
79 Instance: OISM_VRF Family: INET6

```

The below CLI output shows the Type 3 routes on the SL, which is used for handling the BUM traffic. The Ingress replication routes which are critical in building the tunnels are highlighted in the output below. SL1 (R0)

```

1 R0-SL1> show route table bgp.evpn.0 match-prefix 3:* extensive
2
3 bgp.evpn.0: 109 destinations, 109 routes (109 active, 0 holddown, 0 hidden)
4 3:1.1.1.1::100::1.1.1.1/248 IM (1 entry, 1 announced)
5 TSI:
6 Page 0 idx 0, (group ibgp type Internal) Type 1 val 0x55bff6fe1078 (adv_entry)

```

```

 7   Advertised metrics:
 8   Flags: Nexthop Change
 9   Nexthop: Self
10   Localpref: 100
11   AS path: [64512] I
12   Communities: target:1:1 encapsulation:vxlan(0x8) evpn-mcast-flags:0x9:igmp-snooping-enabled:
      oism
13   PMSI: Flags 0x0: Label 6: Type INGRESS-REPLICATION 1.1.1.1
14   Advertise: 00000007
15   Path 3:1.1.1.1:1::100::1.1.1.1
16   Vector len 4. Val: 0
17   *EVPN Preference: 170
18   Next hop type: Indirect, Next hop index: 0
19   Address: 0x55bff66a575c
20   Next-hop reference count: 78, key opaque handle: (nil), non-key opaque handle: (nil)
21   Protocol next hop: 1.1.1.1
22   Indirect next hop: (nil) - INH Session ID: 0
23   State: <Secondary Active Int Ext>
24   Age: 1w5d 18:39:28
25   Validation State: unverified
26   Task: EVPN_OISM-evpn
27   Announcement bits (1): 0-BGP_RT_Background
28   AS path: I
29   Communities: target:1:1 encapsulation:vxlan(0x8) evpn-mcast-flags:0x9:igmp-snooping-
      enabled:oism
30   Route Label: 100
31   PMSI: Flags 0x0: Label 100: Type INGRESS-REPLICATION 1.1.1.1
32   Primary Routing Table: EVPN_OISM.evpn.0
33   Thread: junos-main
34
35   3:1.1.1.1:1::101::1.1.1.1/248 IM (1 entry, 1 announced)
36   TSI:
37   Page 0 idx 0, (group ibgp type Internal) Type 1 val 0x55bff6fe10b0 (adv_entry)
38   Advertised metrics:
39   Flags: Nexthop Change
40   Nexthop: Self
41   Localpref: 100
42   AS path: [64512] I
43   Communities: target:1:1 encapsulation:vxlan(0x8) evpn-mcast-flags:0x9:igmp-snooping-enabled:
      oism
44   PMSI: Flags 0x0: Label 6: Type INGRESS-REPLICATION 1.1.1.1
45   Advertise: 00000007
46   Path 3:1.1.1.1:1::101::1.1.1.1
47   Vector len 4. Val: 0
48   *EVPN Preference: 170
49   Next hop type: Indirect, Next hop index: 0
50   Address: 0x55bff66a575c
51   Next-hop reference count: 78, key opaque handle: (nil), non-key opaque handle: (nil)
52   Protocol next hop: 1.1.1.1
53   Indirect next hop: (nil) - INH Session ID: 0
54   State: <Secondary Active Int Ext>
55   Age: 1w5d 18:39:28
56   Validation State: unverified
57   Task: EVPN_OISM-evpn
58   Announcement bits (1): 0-BGP_RT_Background
59   AS path: I
60   Communities: target:1:1 encapsulation:vxlan(0x8) evpn-mcast-flags:0x9:igmp-snooping-
      enabled:oism
61   Route Label: 101
62   PMSI: Flags 0x0: Label 101: Type INGRESS-REPLICATION 1.1.1.1
63   Primary Routing Table: EVPN_OISM.evpn.0
64   Thread: junos-main
65   3:1.1.1.1:1::900::1.1.1.1/248 IM (1 entry, 1 announced)
66   TSI:
67   Page 0 idx 0, (group ibgp type Internal) Type 1 val 0x55bff6fe4178 (adv_entry)
68   Advertised metrics:
69   Flags: Nexthop Change
70   Nexthop: Self
71   Localpref: 100
72   AS path: [64512] I
73   Communities: target:1:1 encapsulation:vxlan(0x8) evpn-mcast-flags:0x9:igmp-snooping-enabled:
      oism
74   PMSI: Flags 0x0: Label 56: Type INGRESS-REPLICATION 1.1.1.1
75   Advertise: 00000007

```

SL2 (R1)

```

1  R1-SL2> show route table bgp.evpn.0 match-prefix 3:* extensive
2
3  bgp.evpn.0: 109 destinations, 109 routes (109 active, 0 holddown, 0 hidden)

```

```

4 3:1.1.1.2:1::100::1.1.1.2/248 IM (1 entry, 1 announced)
5 TSI:
6 Page 0 idx 0, (group ibgp type Internal) Type 1 val 0x56110ebdc428 (adv_entry)
7   Advertised metrics:
8     Flags: Nexthop Change
9     Nexthop: Self
10    Localpref: 100
11    AS path: [64512] I
12    Communities: target:1:1 encapsulation:vxlan(0x8) evpn-mcast-flags:0x9:igmp-snooping-enabled:
13                  oism
14    PMSI: Flags 0x0: Label 6: Type INGRESS-REPLICATION 1.1.1.2
15    Advertise: 00000007
16    Path 3:1.1.1.2:1::100::1.1.1.2
17    Vector len 4. Val: 0
18      *EVPN Preference: 170
19        Next hop type: Indirect, Next hop index: 0
20        Address: 0x56110e28c1dc
21        Next-hop reference count: 60, key opaque handle: (nil), non-key opaque handle: (nil)
22        Protocol next hop: 1.1.1.2
23        Indirect next hop: (nil) - INH Session ID: 0
24        State: <Secondary Active Int Ext>
25        Age: 1w5d 18:41:03
26        Validation State: unverified
27        Task: EVPN_OISM-evpn
28        Announcement bits (1): 0-BGP_RT_Background
29        AS path: I
30        Communities: target:1:1 encapsulation:vxlan(0x8) evpn-mcast-flags:0x9:igmp-snooping-
31                      enabled:oism
32        Route Label: 100
33        PMSI: Flags 0x0: Label 100: Type INGRESS-REPLICATION 1.1.1.2
34        Primary Routing Table: EVPN_OISM.evpn.0
35        Thread: junos-main
36 3:1.1.1.2:1::101::1.1.1.2/248 IM (1 entry, 1 announced)
37 TSI:
38 Page 0 idx 0, (group ibgp type Internal) Type 1 val 0x56110ebdc690 (adv_entry)
39   Advertised metrics:
40     Flags: Nexthop Change
41     Nexthop: Self
42     Localpref: 100
43     AS path: [64512] I
44     Communities: target:1:1 encapsulation:vxlan(0x8) evpn-mcast-flags:0x9:igmp-snooping-enabled:
45                   oism
46     PMSI: Flags 0x0: Label 6: Type INGRESS-REPLICATION 1.1.1.2
47     Advertise: 00000007
48     Path 3:1.1.1.2:1::101::1.1.1.2
49     Vector len 4. Val: 0
50       *EVPN Preference: 170
51         Next hop type: Indirect, Next hop index: 0
52         Address: 0x56110e28c1dc
53         Next-hop reference count: 60, key opaque handle: (nil), non-key opaque handle: (nil)
54         Protocol next hop: 1.1.1.2
55         Indirect next hop: (nil) - INH Session ID: 0
56         State: <Secondary Active Int Ext>
57         Age: 1w5d 18:41:03
58         Validation State: unverified
59         Task: EVPN_OISM-evpn
60         Announcement bits (1): 0-BGP_RT_Background
61         AS path: I
62         Communities: target:1:1 encapsulation:vxlan(0x8) evpn-mcast-flags:0x9:igmp-snooping-
63                       enabled:oism
64         Route Label: 101
65         PMSI: Flags 0x0: Label 101: Type INGRESS-REPLICATION 1.1.1.2
66         Primary Routing Table: EVPN_OISM.evpn.0
67         Thread: junos-main
68 3:1.1.1.2:1::900::1.1.1.2/248 IM (1 entry, 1 announced)
69 TSI:
70 Page 0 idx 0, (group ibgp type Internal) Type 1 val 0x56110ebddd50 (adv_entry)
71   Advertised metrics:
72     Flags: Nexthop Change
73     Nexthop: Self
74     Localpref: 100
75     AS path: [64512] I
76     Communities: target:1:1 encapsulation:vxlan(0x8) evpn-mcast-flags:0x9:igmp-snooping-enabled:
77                   oism
78     PMSI: Flags 0x0: Label 56: Type INGRESS-REPLICATION 1.1.1.2
79     Advertise: 00000007

```

The below CLI output show the Type 6 routes on the SLs generated locally or learnt from remote leaves.

SL1 (R0)

```

1 R0-SL1> show route table bgp.evpn.0 match-prefix 6*
2 bgp.evpn.0: 106 destinations, 106 routes (106 active, 0 holddown, 0 hidden)
3 + = Active Route, - = Last Active, * = Both
4
5 6:1.1.1.1:1::900::225.1.1.10::1.1.1.1/520
6      *[EVPN/170] 23:23:45
7      Indirect
8 6:1.1.1.1:1::900::225.1.1.20::1.1.1.1/520
9      *[EVPN/170] 23:23:45
10     Indirect
11 6:1.1.1.2:1::900::225.1.1.10::1.1.1.2/520
12     *[BGP/170] 23:10:07, localpref 100, from 1.1.1.2
13     AS path: I, validation-state: unverified
14     > to 10.1.1.1 via et-0/0/7.0
15     to 11.1.1.1 via et-0/0/9.0
16 6:1.1.1.2:1::900::225.1.1.20::1.1.1.2/520
17     *[BGP/170] 23:09:38, localpref 100, from 1.1.1.2
18     AS path: I, validation-state: unverified
19     to 10.1.1.1 via et-0/0/7.0
20     > to 11.1.1.1 via et-0/0/9.0
21 6:1.1.1.5:1::100::0.0.0.0::1.1.1.5/520
22     *[BGP/170] 1d 21:44:08, localpref 100, from 1.1.1.5
23     AS path: I, validation-state: unverified
24     to 10.1.1.1 via et-0/0/7.0
25     > to 11.1.1.1 via et-0/0/9.0
26 6:1.1.1.5:1::101::0.0.0.0::1.1.1.5/520
27     *[BGP/170] 1d 21:44:08, localpref 100, from 1.1.1.5
28     AS path: I, validation-state: unverified
29     to 10.1.1.1 via et-0/0/7.0
30     > to 11.1.1.1 via et-0/0/9.0
31 6:1.1.1.6:1::100::0.0.0.0::1.1.1.6/520
32     *[BGP/170] 1d 21:46:50, localpref 100, from 1.1.1.6
33     AS path: I, validation-state: unverified
34     to 10.1.1.1 via et-0/0/7.0
35     > to 11.1.1.1 via et-0/0/9.0
36 6:1.1.1.6:1::101::0.0.0.0::1.1.1.6/520
37     *[BGP/170] 1d 21:46:50, localpref 100, from 1.1.1.6
38     AS path: I, validation-state: unverified
39     to 10.1.1.1 via et-0/0/7.0
40     > to 11.1.1.1 via et-0/0/9.0

```

SL2 (R1)

```

1 R1-SL2> show route table bgp.evpn.0 match-prefix 6*
2 bgp.evpn.0: 106 destinations, 106 routes (106 active, 0 holddown, 0 hidden)
3 + = Active Route, - = Last Active, * = Both
4
5 6:1.1.1.1:1::900::225.1.1.10::1.1.1.1/520
6      *[BGP/170] 23:27:25, localpref 100, from 1.1.1.1
7      AS path: I, validation-state: unverified
8      > to 12.1.1.1 via et-0/0/1.0
9      to 13.1.1.1 via et-0/0/7.0
10 6:1.1.1.1:1::900::225.1.1.20::1.1.1.1/520
11     *[BGP/170] 23:27:25, localpref 100, from 1.1.1.1
12     AS path: I, validation-state: unverified
13     to 12.1.1.1 via et-0/0/1.0
14     > to 13.1.1.1 via et-0/0/7.0
15 6:1.1.1.2:1::900::225.1.1.10::1.1.1.2/520
16     *[EVPN/170] 23:13:47
17     Indirect
18 6:1.1.1.2:1::900::225.1.1.20::1.1.1.2/520
19     *[EVPN/170] 23:13:18
20     Indirect
21 6:1.1.1.5:1::100::0.0.0.0::1.1.1.5/520
22     *[BGP/170] 23:27:22, localpref 100, from 1.1.1.5
23     AS path: I, validation-state: unverified
24     to 12.1.1.1 via et-0/0/1.0
25     > to 13.1.1.1 via et-0/0/7.0
26 6:1.1.1.5:1::101::0.0.0.0::1.1.1.5/520
27     *[BGP/170] 23:27:22, localpref 100, from 1.1.1.5
28     AS path: I, validation-state: unverified
29     to 12.1.1.1 via et-0/0/1.0
30     > to 13.1.1.1 via et-0/0/7.0
31 6:1.1.1.6:1::100::0.0.0.0::1.1.1.6/520
32     *[BGP/170] 23:27:20, localpref 100, from 1.1.1.6
33     AS path: I, validation-state: unverified
34     to 12.1.1.1 via et-0/0/1.0

```

```

35      > to 13.1.1.1 via et-0/0/7.0
36 6:1.1.1.6:1::101::0.0.0.0::1.1.1.6/520
37      *[BGP/170] 23:27:20, localpref 100, from 1.1.1.6
38      AS path: I, validation-state: unverified
39      to 12.1.1.1 via et-0/0/1.0
40      > to 13.1.1.1 via et-0/0/7.0

```

The below CLI output show the Type 7 routes on the SLs generated locally or learnt from remote leaves.

SL1 (R0)

```

1 R0-SL1> show route table bgp.evpn.0 match-prefix 7*
2 bgp.evpn.0: 109 destinations, 109 routes (109 active, 0 holddown, 0 hidden)
3 + = Active Route, - = Last Active, * = Both
4
5 7:1.1.1.1:1::1111111111111111::100::225.1.1.10::1.1.1.1/600
6      *[EVPN/170] 00:07:54
7      Indirect
8 7:1.1.1.1:1::1111111111111111::101::225.1.1.20::1.1.1.1/600
9      *[EVPN/170] 00:07:54
10     Indirect

```

SL2 (R1)

```

1 R1-SL2> show route table bgp.evpn.0 match-prefix 7*
2 bgp.evpn.0: 109 destinations, 109 routes (109 active, 0 holddown, 0 hidden)
3 + = Active Route, - = Last Active, * = Both
4
5 7:1.1.1.1:1::1111111111111111::100::225.1.1.10::1.1.1.1/600
6      *[BGP/170] 00:08:08, localpref 100, from 1.1.1.1
7      AS path: I, validation-state: unverified
8      to 12.1.1.1 via et-0/0/1.0
9      > to 13.1.1.1 via et-0/0/7.0
10 7:1.1.1.1:1::1111111111111111::101::225.1.1.20::1.1.1.1/600
11      *[BGP/170] 00:08:07, localpref 100, from 1.1.1.1
12      AS path: I, validation-state: unverified
13      > to 12.1.1.1 via et-0/0/1.0
14      to 13.1.1.1 via et-0/0/7.0

```

The below CLI output show the EVPN database and membership details on the SLs. Also it displays the multicast next-hops with its respective VTEP tunnels.

SL1 (R0)

```

1 R0-SL1> show evpn igmp-snooping database extensive group 225.1.1.10
2 Instance: EVPN_OISM
3   VN Identifier: 100
4   Group IP: 225.1.1.10, Source IP: 0.0.0.0
5   Access OIF Count: 1
6   Interface ESI Local Remote
7   ae1.100 00:11:11:11:11:11:11:11:11:11 1 0
8
9 R0-SL1> show evpn igmp-snooping database extensive group 225.1.1.20
10 Instance: EVPN_OISM
11   VN Identifier: 101
12   Group IP: 225.1.1.20, Source IP: 0.0.0.0
13   Access OIF Count: 1
14   Interface ESI Local Remote
15   ae1.101 00:11:11:11:11:11:11:11:11:11 1 0
16
17 R0-SL1> show evpn igmp-snooping database extensive l2-domain-id 100
18 Instance: EVPN_OISM
19   VN Identifier: 100
20   Group IP: 225.1.1.10, Source IP: 0.0.0.0
21   Access OIF Count: 1
22   Interface ESI Local Remote
23   ae1.100 00:11:11:11:11:11:11:11:11:11 1 0
24
25 R0-SL1> show evpn igmp-snooping database extensive l2-domain-id 101
26 Instance: EVPN_OISM
27   VN Identifier: 101
28   Group IP: 225.1.1.20, Source IP: 0.0.0.0
29   Access OIF Count: 1
30   Interface ESI Local Remote
31   ae1.101 00:11:11:11:11:11:11:11:11:11 1 0
32

```

```

33 R0-SL1> show evpn multicast-snooping next-hops
34 Family: INET
35 ID Refcount KRefCount Downstream interface Addr
36 47640 21 6 vtep.32770
37                                     vtep.32772
38                                     vtep.32771
39 47009 3 1 vtep.32770
40                                     vtep.32772
41                                     vtep.32771
42 47014 3 1
43 47018 3 1 vtep.32770
44                                     vtep.32772
45 47025 3 1
46 47001 3 1 vtep.32770
47                                     vtep.32772
48                                     vtep.32771
49 47003 3 1
50 47005 3 1 vtep.32770
51                                     vtep.32772
52 47007 3 1
53 47296 3 1 vtep.32770
54                                     vtep.32772
55                                     vtep.32771
56 47298 3 1
57 47300 3 1 vtep.32770
58                                     vtep.32772
59 47302 3 1

```

SL2 (R1)

```

1 R1-SL2> show evpn igmp-snooping database extensive l2-domain-id 100
2 Instance: EVPN_OISM
3 VN Identifier: 100
4 Group IP: 225.1.1.10, Source IP: 0.0.0.0
5 Access OIF Count: 1
6 Interface ESI Local Remote
7 ae1.100 00:11:11:11:11:11:11:11:11:11 0 1
8
9 R1-SL2> show evpn igmp-snooping database extensive l2-domain-id 101
10 Instance: EVPN_OISM
11 VN Identifier: 101
12 Group IP: 225.1.1.20, Source IP: 0.0.0.0
13 Access OIF Count: 1
14 Interface ESI Local Remote
15 ae1.101 00:11:11:11:11:11:11:11:11:11 0 1
16
17 R1-SL2> show evpn igmp-snooping database extensive group 225.1.1.10
18 Instance: EVPN_OISM
19 VN Identifier: 100
20 Group IP: 225.1.1.10, Source IP: 0.0.0.0
21 Access OIF Count: 1
22 Interface ESI Local Remote
23 ae1.100 00:11:11:11:11:11:11:11:11:11 0 1
24 R1-SL2> show evpn igmp-snooping database extensive group 225.1.1.20
25 Instance: EVPN_OISM
26 VN Identifier: 101
27 Group IP: 225.1.1.20, Source IP: 0.0.0.0
28 Access OIF Count: 1
29 Interface ESI Local Remote
30 ae1.101 00:11:11:11:11:11:11:11:11:11 0 1
31 R1-SL2> show evpn multicast-snooping next-hops
32 Family: INET
33 ID Refcount KRefCount Downstream interface Addr
34 47605 21 6 vtep.32770
35                                     vtep.32771
36                                     vtep.32772
37 47475 3 1 vtep.32770
38                                     vtep.32771
39                                     vtep.32772
40 47477 3 1
41 47479 3 1 vtep.32771
42                                     vtep.32772
43 47481 3 1
44 47483 3 1 vtep.32770
45                                     vtep.32771
46                                     vtep.32772
47 47485 3 1
48 47487 3 1 vtep.32771
49                                     vtep.32772
50 47489 3 1

```

```

51 47467 3 1 vtep.32770
52          vtep.32771
53          vtep.32772
54 47469 3 1
55 47471 3 1 vtep.32771
56          vtep.32772
57 47473 3 1

```

In EVPN table, the 0.0 mesh group is for CE mesh group and 0.1 mesh group is for VE mesh group. Whereas, the MCSNOOPD table it is other way around 0.0 mesh group is for VE and 0.1 is for CE. The below output taken from the SL1, SL2 shows these routes entries to indicate the routes along with VTEP tunnel association.

The routes in EVPN_OISM.evpn-mcsn.1 are EVPN-MCSN family routes which will be communicated from RPD to MCSNOOPD. These are not used in forwarding and for internal use. These routes and next-hops are used in MCSNOOPD to install routes. The MCSNOOPD routes are ones used for forwarding the multicast traffic.

SL1 (R0)

```

1 R0-SL1> show route table EVPN_OISM.evpn-mcsn.1
2 EVPN_OISM.evpn-mcsn.1: 24 destinations, 24 routes (24 active, 0 holddown, 0 hidden)
3 + = Active Route, - = Last Active, * = Both
4
5 0.3,0.0,0.0/48 *[Multicast/180] 6d 19:05:14
6     to 1.1.1.5 via vtep.32770
7     to 1.1.1.6 via vtep.32772
8     to 1.1.1.2 via vtep.32771
9 0.3,0.0,0.0,224.0.0.0/52*[Multicast/180] 6d 19:05:14
10    to 1.1.1.5 via vtep.32770
11    to 1.1.1.6 via vtep.32772
12 0.3,0.0,0.0,225.1.1.10/80*[Multicast/180] 6d 18:51:36
13    to 1.1.1.5 via vtep.32770
14    to 1.1.1.6 via vtep.32772
15    to 1.1.1.2 via vtep.32771
16 0.3,0.0,0.0,225.1.1.20/80*[Multicast/180] 6d 18:51:07
17    to 1.1.1.5 via vtep.32770
18    to 1.1.1.6 via vtep.32772
19    to 1.1.1.2 via vtep.32771
20 0.3,0.1,0.0/48 *[Multicast/180] 6d 19:05:14
21 0.3,0.1,0.0,224.0.0.0/52*[Multicast/180] 6d 19:05:14
22 0.3,0.1,0.0,225.1.1.10/80*[Multicast/180] 6d 18:51:36
23     Multicast (IPv4) Composite
24 0.3,0.1,0.0,225.1.1.20/80*[Multicast/180] 6d 18:51:07
25     Multicast (IPv4) Composite
26 0.4,0.0,0.0/48 *[Multicast/180] 6d 19:05:14
27    to 1.1.1.5 via vtep.32770
28    to 1.1.1.6 via vtep.32772
29    to 1.1.1.2 via vtep.32771
30 0.4,0.0,0.0,224.0.0.0/52*[Multicast/180] 6d 19:05:14
31    to 1.1.1.5 via vtep.32770
32    to 1.1.1.6 via vtep.32772
33 0.4,0.0,0.0,225.1.1.10/80*[Multicast/180] 6d 18:51:36
34    to 1.1.1.5 via vtep.32770
35    to 1.1.1.6 via vtep.32772
36    to 1.1.1.2 via vtep.32771
37 0.4,0.0,0.0,225.1.1.20/80*[Multicast/180] 6d 18:51:07
38    to 1.1.1.5 via vtep.32770
39    to 1.1.1.6 via vtep.32772
40    to 1.1.1.2 via vtep.32771
41 0.4,0.1,0.0/48 *[Multicast/180] 6d 19:05:14
42 0.4,0.1,0.0,224.0.0.0/52*[Multicast/180] 6d 19:05:14
43 0.4,0.1,0.0,225.1.1.10/80*[Multicast/180] 6d 18:51:36
44     Multicast (IPv4) Composite
45 0.4,0.1,0.0,225.1.1.20/80*[Multicast/180] 6d 18:51:07
46     Multicast (IPv4) Composite
47 0.5,0.0,0.0/48 *[Multicast/180] 4d 20:06:54
48    to 1.1.1.5 via vtep.32770
49    to 1.1.1.6 via vtep.32772
50    to 1.1.1.2 via vtep.32771
51 0.5,0.0,0.0,224.0.0.0/52*[Multicast/180] 4d 20:06:54
52    to 1.1.1.5 via vtep.32770
53    to 1.1.1.6 via vtep.32772
54 0.5,0.0,0.0,225.1.1.10/80*[Multicast/180] 6d 17:15:14
55    to 1.1.1.5 via vtep.32770
56    to 1.1.1.6 via vtep.32772
57    to 1.1.1.2 via vtep.32771
58 0.5,0.0,0.0,225.1.1.20/80*[Multicast/180] 6d 17:15:14

```

```

59         to 1.1.1.5 via vtep.32770
60         to 1.1.1.6 via vtep.32772
61         to 1.1.1.2 via vtep.32771
62 0.5,0.1,0.0/48 *[Multicast/180] 4d 20:06:54
63 0.5,0.1,0.0,224.0.0.0/52*[Multicast/180] 4d 20:06:54
64 0.5,0.1,0.0,225.1.1.10/80*[Multicast/180] 6d 18:51:36
65     Multicast (IPv4) Composite
66 0.5,0.1,0.0,225.1.1.20/80*[Multicast/180] 6d 18:51:07
67     Multicast (IPv4) Composite

```

SL2 (R1)

```

1  R1-SL2> show route table EVPN_OISM.evpn-mcsn.1
2  EVPN_OISM.evpn-mcsn.1: 24 destinations, 24 routes (24 active, 0 holddown, 0 hidden)
3  + = Active Route, - = Last Active, * = Both
4
5  0.7,0.0,0.0/48 *[Multicast/180] 6d 19:08:03
6         to 1.1.1.1 via vtep.32770
7         to 1.1.1.5 via vtep.32771
8         to 1.1.1.6 via vtep.32772
9  0.7,0.0,0.0,224.0.0.0/52*[Multicast/180] 6d 19:08:03
10        to 1.1.1.5 via vtep.32771
11        to 1.1.1.6 via vtep.32772
12 0.7,0.0,0.0,225.1.1.10/80*[Multicast/180] 6d 19:08:03
13        to 1.1.1.1 via vtep.32770
14        to 1.1.1.5 via vtep.32771
15        to 1.1.1.6 via vtep.32772
16 0.7,0.0,0.0,225.1.1.20/80*[Multicast/180] 6d 19:08:03
17        to 1.1.1.1 via vtep.32770
18        to 1.1.1.5 via vtep.32771
19        to 1.1.1.6 via vtep.32772
20 0.7,0.1,0.0/48 *[Multicast/180] 6d 19:08:03
21 0.7,0.1,0.0,224.0.0.0/52*[Multicast/180] 6d 19:08:03
22 0.7,0.1,0.0,225.1.1.10/80*[Multicast/180] 6d 19:08:08
23     Multicast (IPv4) Composite
24 0.7,0.1,0.0,225.1.1.20/80*[Multicast/180] 6d 19:08:08
25     Multicast (IPv4) Composite
26 0.8,0.0,0.0/48 *[Multicast/180] 6d 19:08:03
27        to 1.1.1.1 via vtep.32770
28        to 1.1.1.5 via vtep.32771
29        to 1.1.1.6 via vtep.32772
30 0.8,0.0,0.0,224.0.0.0/52*[Multicast/180] 6d 19:08:03
31        to 1.1.1.5 via vtep.32771
32        to 1.1.1.6 via vtep.32772
33 0.8,0.0,0.0,225.1.1.10/80*[Multicast/180] 6d 19:08:03
34        to 1.1.1.1 via vtep.32770
35        to 1.1.1.5 via vtep.32771
36        to 1.1.1.6 via vtep.32772
37 0.8,0.0,0.0,225.1.1.20/80*[Multicast/180] 6d 19:08:03
38        to 1.1.1.1 via vtep.32770
39        to 1.1.1.5 via vtep.32771
40        to 1.1.1.6 via vtep.32772
41 0.8,0.1,0.0/48 *[Multicast/180] 6d 19:08:03
42 0.8,0.1,0.0,224.0.0.0/52*[Multicast/180] 6d 19:08:03
43 0.8,0.1,0.0,225.1.1.10/80*[Multicast/180] 6d 19:08:08
44     Multicast (IPv4) Composite
45 0.8,0.1,0.0,225.1.1.20/80*[Multicast/180] 6d 19:08:08
46     Multicast (IPv4) Composite
47 0.9,0.0,0.0/48 *[Multicast/180] 4d 20:09:48
48        to 1.1.1.1 via vtep.32770
49        to 1.1.1.5 via vtep.32771
50        to 1.1.1.6 via vtep.32772
51 0.9,0.0,0.0,224.0.0.0/52*[Multicast/180] 4d 20:09:48
52        to 1.1.1.5 via vtep.32771
53        to 1.1.1.6 via vtep.32772
54 0.9,0.0,0.0,225.1.1.10/80*[Multicast/180] 6d 17:18:08
55        to 1.1.1.1 via vtep.32770
56        to 1.1.1.5 via vtep.32771
57        to 1.1.1.6 via vtep.32772
58 0.9,0.0,0.0,225.1.1.20/80*[Multicast/180] 6d 17:18:08
59        to 1.1.1.1 via vtep.32770
60        to 1.1.1.5 via vtep.32771
61        to 1.1.1.6 via vtep.32772
62 0.9,0.1,0.0/48 *[Multicast/180] 4d 20:09:48
63 0.9,0.1,0.0,224.0.0.0/52*[Multicast/180] 4d 20:09:48
64 0.9,0.1,0.0,225.1.1.10/80*[Multicast/180] 6d 19:08:08
65     Multicast (IPv4) Composite
66 0.9,0.1,0.0,225.1.1.20/80*[Multicast/180] 6d 19:08:08
67     Multicast (IPv4) Composite

```

The below CLI output shows the routes for mesh-groups `__all_ces__` and `__ves__` created to manage the multicast traffic replication and duplicates that could possibly be created by the leaf nodes. Also the below highlighted are the next hops programmed for the corresponding routes and the vteps tunnels that are created between the SLs and BLs.

SL1 (R0)

```

1 R0-SL1> show multicast snooping route instance EVPN_OISM vlan v100 extensive mesh-group
  __all_ces__
2 Nexthop Bulking: OFF
3
4         Family: INET
5
6 Group: 224.0.0.0/4
7   Source: *
8   Vlan: v100
9   Mesh-group: __all_ces__
10  Downstream interface list:
11    evpn-core-nh -(47021)
12    Statistics: 0 kbps, 0 pps, 2781 packets
13    Next-hop ID: 47033
14    Route state: Active
15    Forwarding state: Forwarding
16    Sensor ID: 4026531846
17
18 Group: 225.1.1.10/32
19   Source: *
20   Vlan: v100
21   Mesh-group: __all_ces__
22   Downstream interface list:
23     evpn-core-nh -(47290) ae1.100 -(85016) et-0/0/3.100 -(85020)
24     Statistics: 0 kbps, 0 pps, 0 packets
25
26 <SNIP>
27 R0-SL1> show multicast snooping route instance EVPN_OISM vlan v100 extensive mesh-group __ves__
28 Nexthop Bulking: OFF
29         Family: INET
30
31 Group: 224.0.0.0/4
32   Source: *
33   Vlan: v100
34   Mesh-group: __ves__
35   Downstream interface list:
36     evpn-core-nh -(47023)
37     Statistics: 0 kbps, 0 pps, 0 packets
38     Next-hop ID: 47035
39     Route state: Active
40     Forwarding state: Forwarding
41     Sensor ID: 4026531847
42
43 Group: 225.1.1.10/32
44   Source: *
45   Vlan: v100
46   Mesh-group: __ves__
47   Downstream interface list:
48     ae1.100 -(85016) et-0/0/3.100 -(85020)
49     Statistics: 0 kbps, 0 pps, 6778 packets
50
51 R0-SL1> show evpn multicast-snooping next-hops 47021
52 Family: INET
53 ID Refcount KRefCount Downstream interface Addr
54 47021 3 1 vtep.32771
55                                vtep.32772
56 R0-SL1> show multicast snooping next-hops 47033
57 Family: INET
58 ID Refcount KRefCount Downstream interface Addr
59 47033 3 1 47021
60
61 R0-SL1> show interfaces terse vtep
62 Interface Admin Link Proto Local Remote
63 vtep up up
64 vtep.32768 up up
65 vtep.32769 up up
66 vtep.32770 up up ethernet-switching
67 vtep.32771 up up ethernet-switching
68 vtep.32772 up up ethernet-switching
69 R0-SL1> show interfaces vtep.32770

```

```

70 Logical interface vtep.32770 (Index 25305) (SNMP ifIndex 549)
71   Flags: Up Encapsulation: Unspecified DF
72   VXLAN Endpoint Type: Shared Remote, VXLAN Endpoint Address: 1.1.1.2, L3 Routing Instance:
73     default
74   Input packets : 7424411
75   Output packets: 7757318
76   Protocol ethernet-switching, MTU: Unlimited
77   Flags: Trunk-Mode
78 R0-SL1> show interfaces vtep.32771
79   Logical interface vtep.32771 (Index 25306) (SNMP ifIndex 550)
80   Flags: Up Encapsulation: Unspecified DF
81   VXLAN Endpoint Type: Shared Remote, VXLAN Endpoint Address: 1.1.1.5, L3 Routing Instance:
82     default
83   Input packets : 3046
84   Output packets: 7784797
85   Protocol ethernet-switching, MTU: Unlimited
86   Flags: Trunk-Mode
87 R0-SL1> show interfaces vtep.32772
88   Logical interface vtep.32772 (Index 25307) (SNMP ifIndex 552)
89   Flags: Up Encapsulation: Unspecified DF
90   VXLAN Endpoint Type: Shared Remote, VXLAN Endpoint Address: 1.1.1.6, L3 Routing Instance:
91     default
92   Input packets : 353646
93   Output packets: 7780217
94   Protocol ethernet-switching, MTU: Unlimited
95   Flags: Trunk-Mode

```

SL2 (R1)

```

1  R1-SL2> show multicast snooping route instance EVPN_OISM vlan v100 extensive mesh-group
2  __all_ces__
3  Nexthop Bulking: OFF
4  Family: INET
5  Group: 224.0.0.0/4
6  Source: *
7  Vlan: v100
8  Mesh-group: __all_ces__
9  Downstream interface list:
10   evpn-core-nh -(47487)
11   Statistics: 0 kbps, 0 pps, 0 packets
12   Next-hop ID: 47525
13   Route state: Active
14   Forwarding state: Forwarding
15   Sensor ID: 4026531843
16
17  Group: 225.1.1.10/32
18  Source: *
19  Vlan: v100
20  Mesh-group: __all_ces__
21  Downstream interface list:
22   evpn-core-nh -(47793) ae1.100 -(83074) et-0/0/5.100 -(83075)
23   Statistics: 6 kbps, 11 pps, 16785552 packets
24
25  <SNIP>
26
27  R1-SL2> show multicast snooping route instance EVPN_OISM vlan v100 extensive mesh-group __ves__
28  Nexthop Bulking: OFF
29
30  Family: INET
31
32  Group: 224.0.0.0/4
33  Source: *
34  Vlan: v100
35  Mesh-group: __ves__
36  Downstream interface list:
37   evpn-core-nh -(47489)
38   Statistics: 0 kbps, 0 pps, 0 packets
39   Next-hop ID: 47527
40   Route state: Active
41   Forwarding state: Forwarding
42   Sensor ID: 4026531842
43  Group: 225.1.1.10/32
44  Source: *
45  Vlan: v100
46  Mesh-group: __ves__
47  Downstream interface list:
48   ae1.100 -(83074) et-0/0/5.100 -(83075)

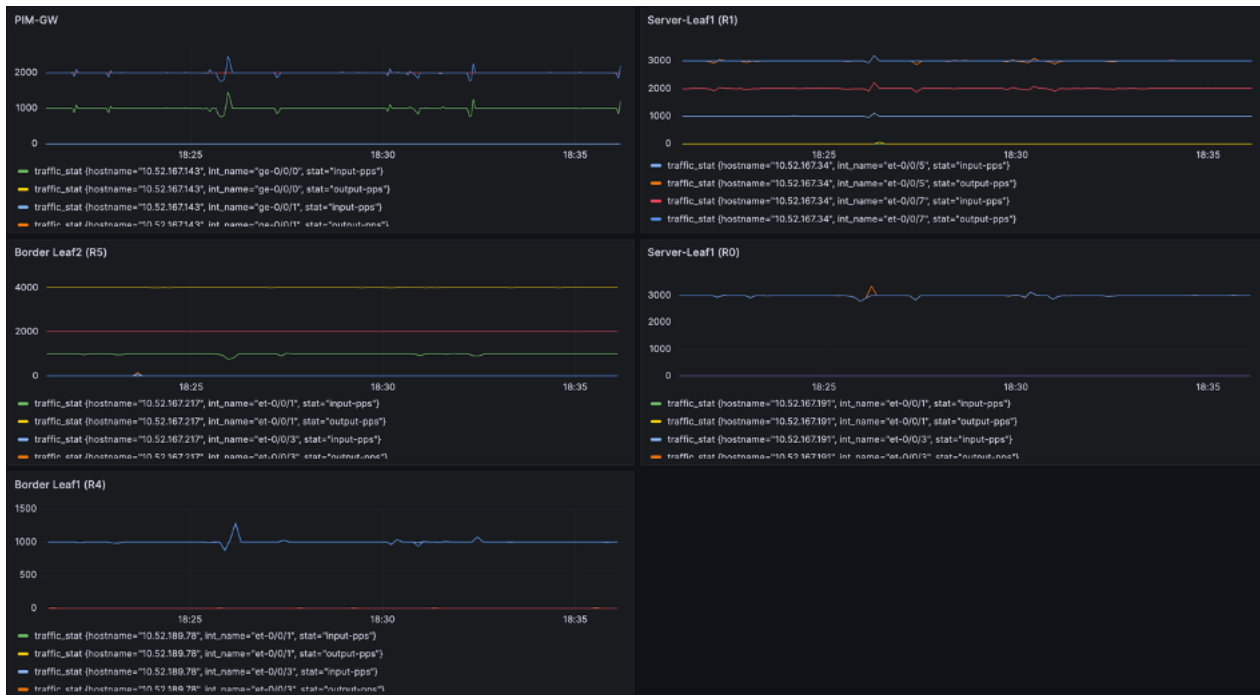
```

```

49      Statistics: 145 kbps, 296 pps, 436299347 packets
50 R1-SL2> show evpn multicast-snooping next-hops 47487
51 Family: INET
52 ID Refcount KRefCount Downstream interface Addr
53 47487 3 1 vtep.32771
54                                vtep.32772
55 R1-SL2> show multicast snooping next-hops 47525
56 Family: INET
57 ID Refcount KRefCount Downstream interface Addr
58 47525 3 1 47487
59 R1-SL2> show interfaces terse vtep
60 Interface Admin Link Proto Local Remote
61 vtep up up
62 vtep.32768 up up
63 vtep.32769 up up
64 vtep.32770 up up ethernet-switching
65 vtep.32771 up up ethernet-switching
66 vtep.32772 up up ethernet-switching
67
68 R1-SL2> show interfaces vtep.32770
69 Logical interface vtep.32770 (Index 19304) (SNMP ifIndex 543)
70 Flags: Up Encapsulation: Unspecified DF
71 VXLAN Endpoint Type: Shared Remote, VXLAN Endpoint Address: 1.1.1.1, L3 Routing Instance:
    default
72 Input packets : 7817082
73 Output packets: 7725299
74 Protocol ethernet-switching, MTU: Unlimited
75 Flags: Trunk-Mode
76
77 R1-SL2> show interfaces vtep.32771
78 Logical interface vtep.32771 (Index 19302) (SNMP ifIndex 548)
79 Flags: Up Encapsulation: Unspecified DF
80 VXLAN Endpoint Type: Shared Remote, VXLAN Endpoint Address: 1.1.1.5, L3 Routing Instance:
    default
81 Input packets : 130092
82 Output packets: 24998805
83 Protocol ethernet-switching, MTU: Unlimited
84 Flags: Trunk-Mode
85
86 R1-SL2> show interfaces vtep.32772
87 Logical interface vtep.32772 (Index 19303) (SNMP ifIndex 549)
88 Flags: Up Encapsulation: Unspecified DF
89 VXLAN Endpoint Type: Shared Remote, VXLAN Endpoint Address: 1.1.1.6, L3 Routing Instance:
    default
90 Input packets : 318336776
91 Output packets: 24989556
92 Protocol ethernet-switching, MTU: Unlimited
93 Flags: Trunk-Mode

```

Traffic Item	Tx Frames	Rx Expected Frames	Rx Frames	Frames Delta	Loss %	Tx Frame Rate	Rx Frame Rate	Tx L1 Rate (bps)	Rx L1 Rate (bps)	Tx Bytes	Tx Rate (bps)	Rx Rate (bps)	Tx Rate (pps)	Rx Rate (pps)
Source-R0-VLAN-100	21,814	65,442	65,272	170	0.260	1,002.253	2,997.760	4,265,587.622	12,758,468.314	33,415,264	513,153.398	1,534,853.000	4,105,227.000	12,278.82
Source-R1-VLAN-101	21,805	65,415	65,414	1	0.002	1,001.608	3,006.323	4,262,842.558	12,794,909.175	33,491,968	512,823.165	1,539,237.000	4,102,585.000	12,313.85
Source-R6-VLAN-1001	21,814	65,442	65,345	97	0.148	1,002.253	3,003.259	4,265,587.622	12,781,870.540	33,456,640	513,153.398	1,537,668.000	4,105,227.000	12,301.34



The show cli commands below highlights the active receivers for the multicast group based on the sources on VLAN100, VLAN101 and the SBD 900. SL1 (R0)

```

1 R0-SL1> show igmp snooping evpn database
2 Instance: EVPN_OISM
3   Bridge-Domain: v100, VN Identifier: 100
4     Group IP: 225.1.1.10, Source IP: 0.0.0.0, Access OIF Count: 2, Core NH: 47021
5     Group IP: 225.1.1.20, Source IP: 0.0.0.0, Access OIF Count: 2, Core NH: 47021
6   Bridge-Domain: v101, VN Identifier: 101
7     Group IP: 225.1.1.20, Source IP: 0.0.0.0, Access OIF Count: 2, Core NH: 47013
8   Bridge-Domain: v900, VN Identifier: 900
9 R0-SL1> show igmp snooping evpn membership
10 Instance: EVPN_OISM
11
12 Vlan: v100, EVPN-Core-NH: 47021
13 Learning-Domain: default
14 Interface: ae1.100, Groups: 2
15   Group: 225.1.1.10
16     Group mode: Exclude
17     Source: 0.0.0.0
18     Type: Local
19   Group: 225.1.1.20
20     Group mode: Exclude
21     Source: 0.0.0.0
22     Type: Local
23 Interface: et-0/0/1.100, Groups: 0
24 Interface: et-0/0/3.100, Groups: 2
25   Group: 225.1.1.10
26     Group mode: Exclude
27     Source: 0.0.0.0
28     Type: Local
29   Group: 225.1.1.20
30     Group mode: Exclude
31     Source: 0.0.0.0
32     Type: Local
33
34 Vlan: v101, EVPN-Core-NH: 47013
35 Learning-Domain: default
36 Interface: ae1.101, Groups: 1
37   Group: 225.1.1.20
38     Group mode: Exclude
39     Source: 0.0.0.0
40     Type: Local
41 Interface: et-0/0/1.101, Groups: 0
42 Interface: et-0/0/3.101, Groups: 1
43   Group: 225.1.1.20
44     Group mode: Exclude
45     Source: 0.0.0.0
46     Type: Local

```

```

47 Vlan: v900, EVPN-Core-NH: 47005
48
49 Learning-Domain: default
50
51 R0-SL1> show multicast route instance OISM_VRF extensive
52 Instance: OISM_VRF Family: INET
53
54 Group: 225.1.1.10
55   Source: 101.1.1.3/32
56   Upstream interface: irb.101
57   Downstream interface list:
58     irb.100 irb.900
59   Number of outgoing interfaces: 2
60   Session description: Unknown
61   Statistics: 546 kbps, 1114 pps, 481531 packets
62   Next-hop ID: 47328
63   Upstream protocol: Multicast
64   Route state: Active
65   Forwarding state: Forwarding
66   Cache lifetime/timeout: 360 seconds
67   Wrong incoming interface notifications: 4
68   Uptime: 00:09:18
69   Sensor ID: 0xf0000007
70
71 Group: 225.1.1.20
72   Source: 100.1.1.1/32
73   Upstream interface: irb.100
74   Downstream interface list:
75     irb.101 irb.900
76   Number of outgoing interfaces: 2
77   Session description: Unknown
78   Statistics: 547 kbps, 1116 pps, 527927 packets
79   Next-hop ID: 47332
80   Upstream protocol: Multicast
81   Route state: Active
82   Forwarding state: Forwarding
83   Cache lifetime/timeout: 360 seconds
84   Wrong incoming interface notifications: 6
85   Uptime: 00:09:18
86   Sensor ID: 0xf0000009
87 Instance: OISM_VRF Family: INET6

```

SL2 (R1)

```

1 R1-SL1> show lldp neighbors
2 Local Interface Parent Interface Chassis Id Port info System Name
3 et-0/0/7 - 0c:68:be:02:03:57 et-0/0/9 R3-re0.englab.juniper.net
4 et-0/0/3 ae1 2c:6b:f5:be:bc:c0 ge-0/0/2 R7-re0.englab.juniper.net
5 re0:mgmt-0 - 50:7c:6f:05:29:f0 tap5 q-pod-106n.englab.juniper.net
6 et-0/0/1 - 70:8f:ei:25:03:57 et-0/0/7 R2-re0.englab.juniper.net
7
8 R1-SL1> show igmp snooping evpn database
9 Instance: EVPN_OISM
10   Bridge-Domain: v100, VN Identifier: 100
11     Group IP: 225.1.1.10, Source IP: 0.0.0.0, Access OIF Count: 2, Core NH: 47487
12     Group IP: 225.1.1.20, Source IP: 0.0.0.0, Access OIF Count: 1, Core NH: 47487
13   Bridge-Domain: v101, VN Identifier: 101
14     Group IP: 225.1.1.20, Source IP: 0.0.0.0, Access OIF Count: 2, Core NH: 47479
15   Bridge-Domain: v900, VN Identifier: 900
16
17 R1-SL1> show igmp snooping evpn membership
18 Instance: EVPN_OISM
19
20 Vlan: v100, EVPN-Core-NH: 47487
21
22 Learning-Domain: default
23 Interface: ae1.100, Groups: 2
24   Group: 225.1.1.10
25     Group mode: Exclude
26     Source: 0.0.0.0
27     Type: Remote
28   Group: 225.1.1.20
29     Group mode: Exclude
30     Source: 0.0.0.0
31     Type: Remote
32 Interface: et-0/0/5.100, Groups: 1
33   Group: 225.1.1.10
34     Group mode: Exclude
35     Source: 0.0.0.0
36     Type: Local

```

The screenshot displays the NetworkMiner application interface. On the left, a sidebar contains a navigation tree with the following structure:

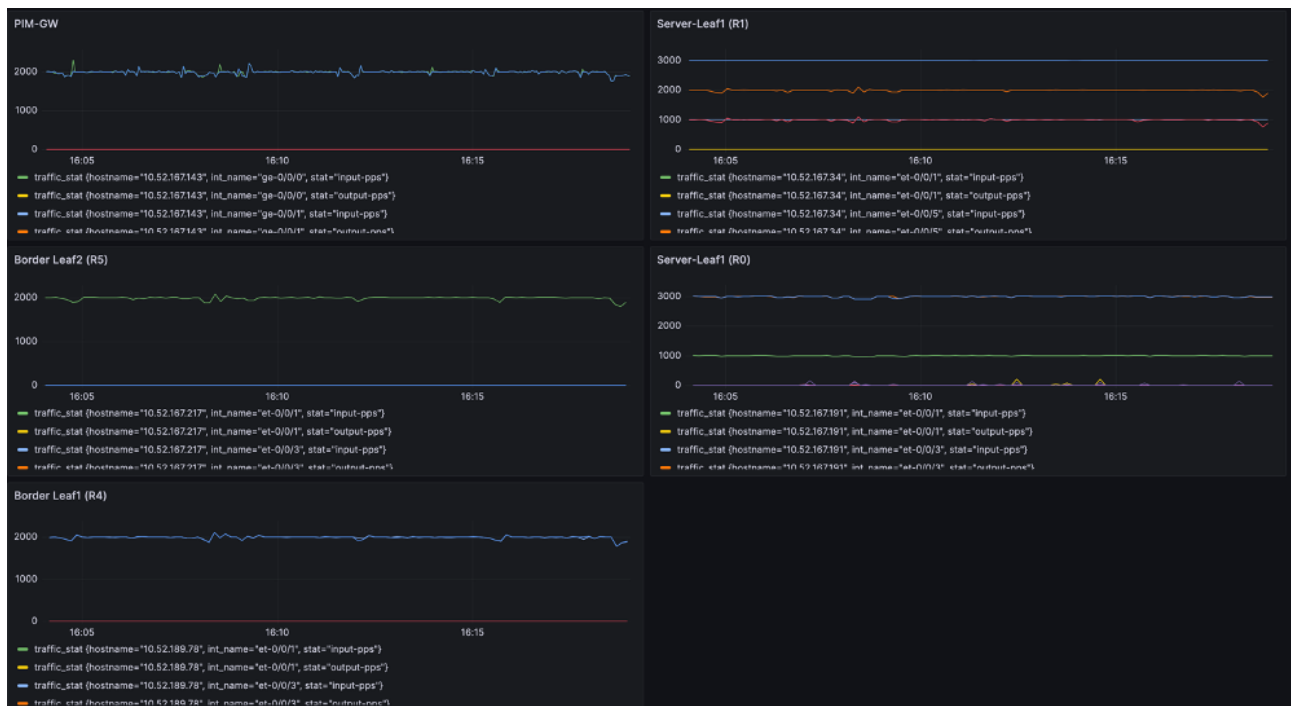
- Overview
 - Scenario
 - Ports
 - Chassis
 - Protocols
 - Protocol Interfaces
 - IGMP
 - R0-et-0/0/3 <=> R9 (ixia2) Running
 - R1-et-0/0/5 <=> R11 (ixia4) Running
 - R6-ge-0/0/2 <=> R12 (ixia5) Running
 - R7-ge-0/0/1 <=> R10 (ixia3) Running
 - Static
 - Traffic
 - L2-L3 Traffic Items**
 - Source-R0-VLAN-100
 - Source-R1-VLAN-101
 - Source-R6-VLAN-1000
 - Source-R6-VLAN-1001
 - L2-L3 Flow Groups

The main window shows the configuration for 'L2-L3 Traffic Items'. It includes a table with the following columns: Transmit State, Traffic Item Name, Enabled, Flow Groups, Tx Ports, Rx Ports, and Endpoint/Encapsulation Sets. The table lists four items:

	Transmit State	Traffic Item Name	Enabled	Flow Groups	Tx Ports	Rx Ports	Endpoint/Encapsulation Sets
1		Source-R6-VLAN-1000	☐	1	1	2	1
2		Source-R0-VLAN-100	☒	1	1	4	1
3		Source-R1-VLAN-101	☒	1	1	2	1
4		Source-R6-VLAN-1001	☐	1	1	2	1

Below the table is a 'Summary' tab. It contains a table with the following columns: Tx Port, Tx Frames, Rx Expected Frames, Rx Frames, Frames Delta, Loss %, Tx Frame Rate, Rx Frame Rate, Tx L1 Rate (bps), Rx L1 Rate (bps), Tx Bytes, Tx Rate (Bps), and Rx Rate (Bps). The table lists four items:

	Tx Port	Tx Frames	Rx Expected Frames	Rx Frames	Frames Delta	Loss %	Tx Frame Rate	Rx Frame Rate	Tx L1 Rate (bps)	Rx L1 Rate (bps)	Tx Bytes	Tx Rate (Bps)	Rx Rate (Bps)
1	R0-et-0/0/3 <=> R9 (ixia2)	13,987	27,974	27,826	148	0.529	1,002.852	2,015.703	4,268,138.279	8,578,830.279	14,246,912	513,460.244	1,032,039...
2	R1-et-0/0/5 <=> R11 (ixia4)	13,987	13,987	13,913	74	0.529	1,002.725	1,008.223	4,267,596.630	4,290,998.207	7,123,456	513,395.083	516,210.311
3	R7-ge-0/0/1 <=> R10 (ixia3)	13,987	27,974	27,832	142	0.508	1,002.445	2,014.385	4,266,405.990	8,573,221.607	14,249,984	513,251.848	1,031,365...
4	R6-ge-0/0/2 <=> R12 (ixia5)	13,987	13,987	13,912	75	0.536	1,003.150	1,008.151	4,269,405.380	4,290,688.557	7,122,944	513,612.677	516,173.059



Multicast Group Leave and Join

Below tests focus on testing the Multicast group leave and joins to demonstrate that the OISM is responsive to the Leave and Join events not just in the local DC domain but also to the traffic all the way to the Multicast source & receivers connected with the PIM-GW. The CLI commands below shows the states of SL1 and SL2 when IXIA simulates Multicast group Leave and Joins respectively.

SL1 (R0)

```

1 R1-SL2> show igmp snooping membership instance EVPN_OISM
2 Instance: EVPN_OISM
3 Vlan: v100
4 Learning-Domain: default
5 Interface: ae1.100, Groups: 0
6 Learning-Domain: default
7 Interface: et-0/0/5.100, Groups: 0
8 Vlan: v101
9 Learning-Domain: default
10 Interface: ae1.101, Groups: 0
11
12 Learning-Domain: default
13 Interface: et-0/0/5.101, Groups: 1
14   Group: 225.1.1.20
15     Group mode: Exclude
16     Source: 0.0.0.0
17     Last reported by: 101.1.1.3
18     Group timeout: 239 Type: Dynamic
19 Vlan: v900
20
21 R1-SL2> show igmp snooping evpn database detail
22 Instance: EVPN_OISM
23   Bridge-Domain: v100, VN Identifier: 100
24     Group IP: 225.1.1.10, Source IP: 0.0.0.0
25     Core NH: 47487
26     Access OIF Count: 1
27       Interface Local Remote
28       ae1.100 0 1
29   Bridge-Domain: v101, VN Identifier: 101
30     Group IP: 225.1.1.20, Source IP: 0.0.0.0
31     Core NH: 47479
32     Access OIF Count: 2
33       Interface Local Remote
34       ae1.101 0 1
35       et-0/0/5.101 1 0
36   Bridge-Domain: v900, VN Identifier: 900

```

SL2 (R1)

```

1 R1-SL2> show igmp snooping membership instance EVPN_OISM
2 Instance: EVPN_OISM
3 Vlan: v100
4 Learning-Domain: default
5 Interface: ae1.100, Groups: 0
6 Learning-Domain: default
7 Interface: et-0/0/5.100, Groups: 1
8   Group: 225.1.1.10
9     Group mode: Exclude
10    Source: 0.0.0.0
11    Last reported by: 100.1.1.3
12    Group timeout: 231 Type: Dynamic
13 Vlan: v101
14
15 Learning-Domain: default
16 Interface: ae1.101, Groups: 0
17 Learning-Domain: default
18 Interface: et-0/0/5.101, Groups: 1
19   Group: 225.1.1.20
20     Group mode: Exclude
21    Source: 0.0.0.0
22    Last reported by: 101.1.1.3
23    Group timeout: 202 Type: Dynamic
24 Vlan: v900
25 Learning-Domain: default
26 Interface: et-0/0/5.900, Groups: 0
27
28 R1-SL2> show igmp snooping evpn database detail
29 Instance: EVPN_OISM
30 Bridge-Domain: v100, VN Identifier: 100
31   Group IP: 225.1.1.10, Source IP: 0.0.0.0
32   Core NH: 47487
33   Access OIF Count: 2
34     Interface Local Remote
35     ae1.100 0 1
36     et-0/0/5.100 1 0
37 Bridge-Domain: v101, VN Identifier: 101
38   Group IP: 225.1.1.20, Source IP: 0.0.0.0
39   Core NH: 47479
40   Access OIF Count: 2
41     Interface Local Remote
42     ae1.101 0 1
43     et-0/0/5.101 1 0
44 Bridge-Domain: v900, VN Identifier: 900

```

Figure 10 below shows the multicast traffic drop and rampup during IGMP leave and Join events on SL1 of the OISM topology. These statistics shown below in Figure 10 or 11 are polled from the routers corresponding interfaces. The IGMP hosts connected from SL1 to Ixia simulate Leave and Join events.



Figure 11 below shows the multicast traffic drop and rampup during IGMP leave and Join events on PIM-GW

of the OISM topology. The IGMP hosts connected from PIM-GW directly to Ixia on vlan 1000 simulate Leave and Join events. This is particularly on multicast group 225.1.1.10.



1.4.5 OISM PIM-GW with M-VLAN

Configuration Overview

In OISM External IRB MVLAN model there is no changes in SL configurations. However, BL and PIM-GW undergoes major additions and modifications that include LAG and MVLAN instance related configurations. Below section will focus on the configuration additions on BL and PIM-GW only and the complete underlay, overlay and access side configuration are covered in the Configuration Snippet section.

BL1:

```

1 R0-BL1# show | compare
2 + chassis {
3 +   aggregated-devices {
4 +     ethernet {
5 +       device-count 10;
6 +     }
7 +   }
8 + }
9 [edit interfaces et-0/0/5]
10 + ether-options {
11 +   802.3ad ae0;
12 + }
13 [edit interfaces]
14 + ae0 {
15 +   flexible-vlan-tagging;
16 +   encapsulation flexible-ethernet-services;
17 +   esi {
18 +     00:22:22:22:22:22:22:22:22;
19 +   all-active;
20 + }
21 + aggregated-ether-options {
22 +   lacp {
23 +     active;
24 +     system-id 00:22:22:22:22:22;
25 +   }
26 + }
27 + unit 2000 {
28 +   encapsulation vlan-bridge;
29 +   vlan-id 2000;
30 + }
31 + }
32 [edit interfaces irb]
33 + unit 2000 {

```

```

34 + family inet {
35 + address 80.1.1.50/24 {
36 + virtual-gateway-address 80.1.1.254;
37 + }
38 + }
39 + }
40 [edit]
41 + policy-options {
42 + prefix-list p1 {
43 + 80.1.1.70/32;
44 + }
45 + policy-statement pol {
46 + term 1 {
47 + from {
48 + prefix-list p1;
49 + }
50 + then accept;
51 + }
52 + }
53 + }
54 [edit routing-instances]
55 + EVPN_MVLAN {
56 + instance-type mac-vrf;
57 + protocols {
58 + igmp-snooping {
59 + vlan v2000 {
60 + proxy;
61 + interface ae0.2000 {
62 + multicast-router-interface;
63 + }
64 + }
65 + }
66 + evpn {
67 + encapsulation vxlan;
68 + default-gateway no-gateway-community;
69 + extended-vni-list 2000;
70 + }
71 + }
72 + vtep-source-interface lo0.0;
73 + service-type vlan-aware;
74 + route-distinguisher 1.1.1.5:3;
75 + vrf-target target:1:3;
76 + vlans {
77 + v2000 {
78 + vlan-id 2000;
79 + interface ae0.2000;
80 + l3-interface irb.2000;
81 + vxlan {
82 + vni 2000;
83 + }
84 + }
85 + }
86 + }
87 [edit routing-instances OISM_VRF protocols evpn oism pim-evpn-gateway]
88 + external-irb irb.2000;
89 [edit routing-instances OISM_VRF protocols ospf area 0.0.0.0]
90 interface irb.900 { ... }
91 + interface irb.2000;
92 - interface et-0/0/5.0;
93 [edit routing-instances OISM_VRF protocols pim]
94 + interface irb.2000 {
95 + accept-join-always-from pol;
96 + distributed-dr;
97 + }
98 [edit routing-instances OISM_VRF]
99 + interface irb.2000;

```

BL2:

```

1 R0-BL2# show | compare
2 + chassis {
3 + aggregated-devices {
4 + ethernet {
5 + device-count 10;
6 + }
7 + }
8 + }
9 [edit interfaces et-0/0/5]
10 + ether-options {
11 + 802.3ad ae0;

```

```

12 + }
13 [edit interfaces]
14 + ae0 {
15 + flexible-vlan-tagging;
16 + encapsulation flexible-ethernet-services;
17 + esi {
18 + 00:22:22:22:22:22:22:22:22;
19 + all-active;
20 + }
21 + aggregated-ether-options {
22 + lacp {
23 + active;
24 + system-id 00:22:22:22:22:22;
25 + }
26 + }
27 + unit 2000 {
28 + encapsulation vlan-bridge;
29 + vlan-id 2000;
30 + }
31 + }
32 [edit interfaces irb]
33 + unit 2000 {
34 + virtual-gateway-accept-data;
35 + family inet {
36 + address 80.1.1.60/24 {
37 + virtual-gateway-address 80.1.1.254;
38 + }
39 + }
40 + }
41 [edit]
42 + policy-options {
43 + prefix-list p1 {
44 + 80.1.1.70/32;
45 + }
46 + policy-statement pol {
47 + term 1 {
48 + from {
49 + prefix-list p1;
50 + }
51 + then accept;
52 + }
53 + }
54 + }
55 [edit routing-instances]
56 + EVPN_MVLAN {
57 + instance-type mac-vrf;
58 + protocols {
59 + igmp-snooping {
60 + vlan v2000 {
61 + proxy;
62 + interface ae0.2000 {
63 + multicast-router-interface;
64 + }
65 + }
66 + }
67 + evpn {
68 + encapsulation vxlan;
69 + default-gateway no-gateway-community;
70 + extended-vni-list 2000;
71 + }
72 + }
73 + vtep-source-interface lo0.0;
74 + service-type vlan-aware;
75 + route-distinguisher 1.1.1.6:3;
76 + vrf-target target:1:3;
77 + vlans {
78 + v2000 {
79 + vlan-id 2000;
80 + interface ae0.2000;
81 + l3-interface irb.2000;
82 + vxlan {
83 + vni 2000;
84 + }
85 + }
86 + }
87 + }
88 [edit routing-instances OISM_VRF protocols evpn oism pim-evpn-gateway]
89 + external-irb irb.2000;
90 [edit routing-instances OISM_VRF protocols ospf area 0.0.0.0]
91 + interface irb.900 { ... }
92 + interface irb.2000;

```

```

93 - interface et-0/0/5.0;
94 [edit routing-instances OISM_VRF protocols pim]
95 + interface irb.2000 {
96 +   accept-join-always-from pol;
97 +   distributed-dr;
98 + }
99 [edit routing-instances OISM_VRF]
100 + interface irb.2000;

```

PIM-GW:

```

1 R0-BL1# regress@R6-re0# show | compare
2 [edit chassis]
3 + aggregated-devices {
4 +   ethernet {
5 +     device-count 10;
6 +   }
7 + }
8 [edit interfaces ge-0/0/0]
9 + gigether-options {
10 +   802.3ad ae0;
11 + }
12 [edit interfaces ge-0/0/0]
13 ! inactive: unit 0 { ... }
14 [edit interfaces ge-0/0/1]
15 + gigether-options {
16 +   802.3ad ae0;
17 + }
18 [edit interfaces]
19 + ae0 {
20 +   flexible-vlan-tagging;
21 +   encapsulation flexible-ethernet-services;
22 +   esi {
23 +     00:22:22:22:22:22:22:22:22;
24 +   all-active;
25 + }
26 + aggregated-ether-options {
27 +   lacp {
28 +     active;
29 +     system-id 00:22:22:22:22:21;
30 +   }
31 + }
32 + unit 2000 {
33 +   encapsulation vlan-bridge;
34 +   vlan-id 2000;
35 + }
36 + }
37 + irb {
38 +   unit 2000 {
39 +     virtual-gateway-accept-data;
40 +     family inet {
41 +       address 80.1.1.70/24 {
42 +         virtual-gateway-address 80.1.1.254;
43 +       }
44 +     }
45 +   }
46 + }
47 [edit protocols pim]
48 + interface irb.2000;
49 [edit]
50 + bridge-domains {
51 +   bd-2000 {
52 +     domain-type bridge;
53 +     vlan-id 2000;
54 +     interface ae0.2000;
55 +     routing-interface irb.2000;
56 +   }
57 + }

```

1.5 Operational Overview

Below section covers the different verification CLI commands to check the status of the multicast forwarding on different nodes in the network including the SL, BL, PIM-GW.

There are two different scenarios of traffic flow:

1. Multicast source resides in the EVPN DC and the traffic flows out to PIM GW externally.
2. Traffic comes into the EVPN DC from the multicast source connected externally on PIM GW.

1.5.1 OISM States on BL & PIM-GW

Case 1. Multicast source is inside the DC fabric

The show cli commands below highlights the states of Multicast on the BL and PIM GW, where the active source resides within the EVPN DC and the receivers are within the DC and also in externally connected.

BL1:

```

1 R4-BL1> show pim join instance OISM_VRF extensive
2
3 Instance: PIM.OISM_VRF Family: INET
4 R = Rendezvous Point Tree, S = Sparse, W = Wildcard
5 Group: 225.1.1.10
6   Source: *
7   RP: 1.1.1.7
8   Flags: sparse,rptree,wildcard
9   Upstream interface: irb.2000
10  Upstream neighbor: 80.1.1.70
11  Upstream state: Join to RP
12  Uptime: 02:13:27
13  Downstream neighbors:
14    Interface: irb.900
15      90.1.1.50 State: Join Flags: SRW Timeout: Infinity
16      Uptime: 02:13:27 Time since last Join: 00:14:28
17    Interface: irb.2000
18      80.1.1.60 State: Join Flags: SRW Timeout: 185
19      Uptime: 00:13:25 Time since last Join: 00:00:25
20  Number of downstream interfaces: 2
21  Number of downstream neighbors: 2
22
23 Group: 225.1.1.10
24   Source: 101.1.1.3
25   Flags: sparse,spt
26   Upstream interface: irb.101
27   Upstream neighbor: Direct
28   Upstream state: Local Source, Prune to RP
29   Keepalive timeout: 0
30   Uptime: 02:13:24
31   Downstream neighbors:
32     Interface: irb.2000
33       80.1.1.60 State: Prune Flags: SR Timeout: 185
34       Uptime: 00:13:25 Time since last Prune: 00:00:25
35       80.1.1.70 State: Join Flags: S Timeout: 195
36       Uptime: 00:13:15 Time since last Join: 00:00:15
37   Number of downstream interfaces: 1
38   Number of downstream neighbors: 2
39
40 <Output truncated for brevity>
41 R4-BL1> show multicast route instance OISM_VRF extensive
42 Instance: OISM_VRF Family: INET
43
44 Group: 225.1.1.10
45   Source: 101.1.1.3/32
46   Upstream interface: irb.101
47   Downstream interface list:
48     irb.2000
49   Number of outgoing interfaces: 1
50   Session description: Unknown
51   Statistics: 577 kBps, 1178 pps, 8071586 packets
52   Next-hop ID: 47749
53   Upstream protocol: Multicast
54   Route state: Active
55   Forwarding state: Forwarding
56   Cache lifetime/timeout: 360 seconds
57   Wrong incoming interface notifications: 3
58   Uptime: 02:14:22
59   Sensor ID: 0xf0000029

```

```

60
61 Instance: OISM_VRF Family: INET6
62 R4-BL1> monitor interface traffic
63
64 R4-re0-re0 Seconds: 6 Time: 09:31:15
65
66 Interface Link Input packets (pps) Output packets (pps)
67
68 ae0 Up 2478851 (3) 1116414997 (1003)
69 dsc Up 0 (0) 0 (0)
70 esi Up 0 (0) 0 (0)
71 et-0/0/1 Up 174182952 (1) 828496 (1)
72 et-0/0/3 Up 989327606 (1002) 1033903 (1)
73 et-0/0/5 Up 2478851 (3) 1116414997 (1003)
74 et-0/0/7 Up 1210788863 (1) 2971320 (3)
75 et-0/0/9 Up 0 (0) 66111 (0)
76 et-0/0/11 Up 0 (0) 66109 (0)
77 et-0/0/13 Up 0 (0) 66109 (0)
78 et-0/0/15 Up 0 (0) 66106 (0)

```

BL2:

```

1 R5-BL2> show pim join instance OISM_VRF extensive
2 Instance: PIM.OISM_VRF Family: INET
3 R = Rendezvous Point Tree, S = Sparse, W = Wildcard
4
5 Group: 225.1.1.10
6   Source: *
7   RP: 1.1.1.7
8   Flags: sparse,rptree,wildcard
9   Upstream interface: irb.2000
10  Upstream neighbor: 80.1.1.70
11  Upstream state: Join to RP
12  Uptime: 02:15:08
13  Downstream neighbors:
14    Interface: irb.900
15      90.1.1.60 State: Join Flags: SRW Timeout: Infinity
16      Uptime: 02:15:08 Time since last Join: 00:16:19
17    Interface: irb.2000
18      80.1.1.50 State: Join Flags: SRW Timeout: 196
19      Uptime: 00:15:14 Time since last Join: 00:00:14
20  Number of downstream interfaces: 2
21  Number of downstream neighbors: 2
22 Group: 225.1.1.10
23   Source: 101.1.1.3
24   Flags: sparse,spt
25   Upstream interface: irb.101
26   Upstream neighbor: Direct
27   Upstream state: Local Source, Prune to RP
28   Keepalive timeout: 0
29   Uptime: 02:15:05
30   Downstream neighbors:
31     Interface: irb.900
32       90.1.1.60 State: Join Flags: S Timeout: Infinity
33       Uptime: 02:15:05 Time since last Join: 02:15:05
34     Interface: irb.2000
35       80.1.1.50 State: Prune Flags: SR Timeout: 196
36       Uptime: 00:15:14 Time since last Prune: 00:00:14
37       80.1.1.70 State: Join Flags: S Timeout: 206
38       Uptime: 00:15:04 Time since last Join: 00:00:04
39   Number of downstream interfaces: 2
40   Number of downstream neighbors: 3
41
42 <Output truncated for brevity>
43 R5-BL2> show multicast route extensive instance OISM_VRF
44 Instance: OISM_VRF Family: INET
45 Group: 225.1.1.10
46   Source: 101.1.1.3/32
47   Upstream interface: irb.101
48   Downstream interface list:
49     irb.900 irb.2000
50   Number of outgoing interfaces: 2
51   Session description: Unknown
52   Statistics: 492 kbps, 1004 pps, 8157643 packets
53   Next-hop ID: 47659
54   Upstream protocol: Multicast
55   Route state: Active
56   Forwarding state: Forwarding
57   Cache lifetime/timeout: 360 seconds
58   Wrong incoming interface notifications: 4

```

```

59     Uptime: 02:15:56
60     Sensor ID: 0xf0000031
61 Instance: OISM_VRF Family: INET6
62
63 R5-BL1> monitor interface traffic
64
65 R5-re0-re0 Seconds: 6 Time: 09:33:51
66
67 Interface Link Input packets (pps) Output packets (pps)
68
69 ae0 Up 1209974498 (1) 397661 (1)
70 dsc Up 0 (0) 0 (0)
71 esi Up 0 (0) 0 (0)
72 et-0/0/1 Up 989262611 (1002) 2420236540 (0)
73 et-0/0/3 Up 174142496 (0) 629740 (0)
74 et-0/0/5 Up 1209974498 (1) 397661 (1)
75 et-0/0/7 Up 2971400 (0) 1210791451 (0)
76 et-0/0/9 Up 0 (0) 66106 (0)
77 et-0/0/11 Up 0 (0) 66107 (0)
78 et-0/0/13 Up 0 (0) 66111 (0)
79 et-0/0/15 Up 0 (0) 66122 (0)

```

PIM-GW:

```

1 R6-PIM-GW> show pim join extensive
2 Instance: PIM.master Family: INET
3 R = Rendezvous Point Tree, S = Sparse, W = Wildcard
4
5 Group: 225.1.1.10
6   Source: *
7   RP: 1.1.1.7
8   Flags: sparse,rptree,wildcard
9   Upstream interface: Local
10  Upstream neighbor: Local
11  Upstream state: Local RP
12  Uptime: 1w4d 08:42:07
13  Downstream neighbors:
14    Interface: irb.2000 (assert winner)
15      80.1.1.50 State: Join Flags: SRW Timeout: 176
16      Uptime: 02:12:34 Time since last Join: 00:00:34
17      80.1.1.60 State: Join Flags: SRW Timeout: 176
18      Uptime: 02:11:34 Time since last Join: 00:00:34
19      Assert Winner: 80.1.1.70 Metric: 0 Pref: 2147483648 Timeout: 55
20    Interface: ge-0/0/2.1001
21      25.1.2.1 State: Join Flags: SRW Timeout: Infinity
22      Uptime: 04:32:06 Time since last Join: 02:12:46
23  Number of downstream interfaces: 2
24  Number of downstream neighbors: 3
25
26 Group: 225.1.1.10
27   Source: 101.1.1.3
28   Flags: sparse,spt
29   Upstream interface: irb.2000
30   Upstream neighbor: 80.1.1.50
31   Upstream state: Local RP, Join to Source, No Prune to RP
32   Keepalive timeout: 311
33   Uptime: 02:12:34
34   Downstream neighbors:
35     Interface: irb.2000 (pruned)
36       80.1.1.50 State: Prune Flags: SR Timeout: 176
37       Uptime: 02:12:34 Time since last Prune: 00:00:34
38       80.1.1.60 State: Prune Flags: SR Timeout: 176
39       Uptime: 02:11:34 Time since last Prune: 00:00:34
40     Interface: ge-0/0/2.1001
41       25.1.2.1 State: Join Flags: S Timeout: Infinity
42       Uptime: 02:12:25 Time since last Join: 02:12:25
43   Number of downstream interfaces: 2
44   Number of downstream neighbors: 3
45
46 <SNIP>
47 R6-PIM-GW> show multicast route extensive
48 Instance: master Family: INET
49
50 Group: 225.1.1.10
51   Source: 101.1.1.3/32
52   Upstream interface: irb.2000
53   Downstream interface list:
54     ge-0/0/2.1001
55   Number of outgoing interfaces: 1
56   Session description: Unknown

```

```

57 Statistics: 491 kbps, 1002 pps, 9148322 packets
58 Next-hop ID: 1048575
59 Upstream protocol: PIM
60 Route state: Active
61 Forwarding state: Forwarding
62 Cache lifetime/timeout: 360 seconds
63 Wrong incoming interface notifications: 0
64 Uptime: 02:32:10
65
66 Instance: master Family: INET6
67
68 R6-PIM-GW> monitor interface traffic
69 R6-re0 Seconds: 16 Time: 09:28:49
70
71 Interface Link Input packets (pps) Output packets (pps)
72 ge-0/0/0 Up 1116101154 (1005) 2403761 (1) >>>>>> Interface connected to BL1 (R4)
73 lc-0/0/0 Up 0 0
74 pfh-0/0/0 Up 0 0
75 ge-0/0/1 Up 303472 (0) 1209892130 (0)
76 ge-0/0/2 Up 1228012918 (0) 1735868562 (1005) >>>>>> Interface connected to IXIA

```

Flow Groups	Transmit State	Suspend	Tx Port	Src/Dst Endpoint Pair	Rx Ports	Flow Group Name	Configured Frame Size	Applied Frame Size	Frame Rate	App
1	✓	✗	R1-et-0/0/5 <-> R1...	R0-et-0/0/3 <-> R9 (Dxia2)...	Source-R1-EndpointSet-1...	Fixed: 512	Fixed: 512	1000 fps	1000 f	

Rx Port	Tx Frames	Rx Expected Frames	Rx Frames	Frames Delta	Loss %	Tx Frame Rate	Rx Frame Rate	Tx L1 Rate (bps)	Rx L1 Rate (bps)	Rx Bytes	Tx Rate (Bps)	Rx Rate (Bps)	Tx Rate (bps)
1 R0-et-0/0/3 <-> R9 (Dxia2)	81,964	81,964	81,889	75	0.092	1,002.502	1,002.002	4,266,649.173	4,264,521.169	41,927,168	513,281.104	513,025.103	4,106,248...
2 R7-ge-0/0/1 <-> R10 (Dxia3)	81,964	81,964	81,879	85	0.104	1,002.510	1,002.510	4,266,682.240	4,266,682.240	41,922,048	513,285.082	513,285.082	4,106,280...
3 R6-ge-0/0/2 <-> R12 (Dxia5)	81,964	81,964	81,875	89	0.109	1,002.808	1,002.308	4,267,949.578	4,265,820.924	41,920,000	513,437.543	513,181.465	4,107,500...

Case 2. External multicast source.

The show cli commands below highlights the states of multicast routes on the BL and PIM GW, where the active source resides within the EVPN DC and some receivers are within the DC and others are connected externally.

BL1:

```

1 R4-BL1> show pim join instance OISM_VRF extensive
2
3 Instance: PIM.OISM_VRF Family: INET
4 R = Rendezvous Point Tree, S = Sparse, W = Wildcard
5 Group: 225.1.1.10
6 Source: *
7 RP: 1.1.1.7
8 Flags: sparse,rptree,wildcard
9 Upstream interface: irb.2000
10 Upstream neighbor: 80.1.1.70
11 Upstream state: Join to RP
12 Uptime: 04:54:11
13 Downstream neighbors:
14 Interface: irb.900
15 90.1.1.50 State: Join Flags: SRW Timeout: Infinity
16 Uptime: 04:54:11 Time since last Join: 02:55:12
17 Interface: irb.2000
18 80.1.1.60 State: Join Flags: SRW Timeout: 201
19 Uptime: 02:54:09 Time since last Join: 00:00:09
20 Number of downstream interfaces: 2
21 Number of downstream neighbors: 2
22 Group: 225.1.1.10
23 Source: 25.1.1.2
24 Flags: sparse,spt
25 Upstream interface: irb.2000
26 Upstream neighbor: 80.1.1.70
27 Upstream state: Join to Source, No Prune to RP
28 Keepalive timeout: 0

```

```

29     Uptime: 00:06:10
30     Downstream neighbors:
31         Interface: irb.2000
32             80.1.1.60 State: Join Flags: S Timeout: 201
33             Uptime: 00:06:10 Time since last Join: 00:00:09
34     Number of downstream interfaces: 1
35     Number of downstream neighbors: 1
36 Group: 225.1.1.10
37     Source: 101.1.1.3
38     Flags: sparse
39     Upstream interface: irb.101
40     Upstream neighbor: Direct
41     Upstream state: Local Source, No Prune to RP
42     Keepalive timeout:
43     Uptime: 04:54:08
44     Downstream neighbors:
45         Interface: irb.2000
46             80.1.1.70 State: Join Flags: S Timeout: 151
47             Uptime: 02:53:59 Time since last Join: 00:00:59
48     Number of downstream interfaces: 1
49     Number of downstream neighbors: 1
50 <SNIP>
51
52 R4-BL1> monitor interface traffic
53
54 R4-re0-re0 Seconds: 886 Time: 09:45:55
55 Interface Link Input packets (pps) Output packets (pps)
56
57 ae0 Up 2663520 (1003) 1117041776 (1)
58 dsc Up 0 (0) 0 (0)
59 esi Up 0 (0) 0 (0)
60 et-0/0/1 Up 174183205 (0) 828814 (0)
61 et-0/0/3 Up 989953751 (0) 1034505 (0)
62 et-0/0/5 Up 2663520 (1003) 1117041776 (1)
63 et-0/0/7 Up 1210973222 (1003) 3155673 (1003)
64 et-0/0/9 Up 0 (0) 66142 (0)
65 et-0/0/11 Up 0 (0) 66141 (0)
66 et-0/0/13 Up 0 (0) 66141 (0)
67 et-0/0/15 Up 0 (0) 66138 (0)

```

BL2:

```

1 R5-BL2> show pim join instance OISM_VRF extensive
2
3 Instance: PIM.OISM_VRF Family: INET
4 R = Rendezvous Point Tree, S = Sparse, W = Wildcard
5
6 Group: 225.1.1.10
7     Source: *
8     RP: 1.1.1.7
9     Flags: sparse,rptree,wildcard
10    Upstream interface: irb.2000
11    Upstream neighbor: 80.1.1.70
12    Upstream state: Join to RP
13    Uptime: 04:52:06
14    Downstream neighbors:
15        Interface: irb.900
16            90.1.1.60 State: Join Flags: SRW Timeout: Infinity
17            Uptime: 04:52:06 Time since last Join: 02:53:17
18        Interface: irb.2000
19            80.1.1.50 State: Join Flags: SRW Timeout: 198
20            Uptime: 02:52:12 Time since last Join: 00:00:12
21    Number of downstream interfaces: 2
22    Number of downstream neighbors: 2
23
24 Group: 225.1.1.10
25     Source: 25.1.1.2
26     Flags: sparse,spt
27     Upstream interface: irb.2000
28     Upstream neighbor: 80.1.1.70
29     Upstream state: Join to Source, No Prune to RP
30     Keepalive timeout: 107
31     Uptime: 00:04:13
32     Downstream neighbors:
33         Interface: irb.900
34             90.1.1.60 State: Join Flags: S Timeout: Infinity
35             Uptime: 00:04:13 Time since last Join: 00:04:13
36         Interface: irb.2000
37             80.1.1.50 State: Join Flags: S Timeout: 198
38             Uptime: 00:04:13 Time since last Join: 00:00:12

```

```

39  Number of downstream interfaces: 2
40  Number of downstream neighbors: 2
41  Group: 225.1.1.10
42  Source: 101.1.1.3
43  Flags: sparse,spt
44  Upstream interface: irb.101
45  Upstream neighbor: Direct
46  Upstream state: Local Source, Prune to RP
47  Keepalive timeout: 0
48  Uptime: 04:52:03
49  Downstream neighbors:
50  Interface: irb.900
51  90.1.1.60 State: Join Flags: S Timeout: Infinity
52  Uptime: 04:52:03 Time since last Join: 04:52:03
53  Interface: irb.2000
54  80.1.1.50 State: Prune Flags: SR Timeout: 198
55  Uptime: 02:52:12 Time since last Prune: 00:00:12
56  80.1.1.70 State: Join Flags: S Timeout: 207
57  Uptime: 02:52:02 Time since last Join: 00:00:02
58  Number of downstream interfaces: 2
59  Number of downstream neighbors: 3
60
61  <SNIP>
62  R5-BL2> monitor interface traffic
63
64  R5-re0-re0 Seconds: 747 Time: 09:46:12
65
66  Interface Link Input packets (pps) Output packets (pps)
67
68  ae0 Up 1209975390 (1) 398595 (1)
69  dsc Up 0 (0) 0 (0)
70  esi Up 0 (0) 0 (0)
71  et-0/0/1 Up 989735767 (1) 2420640210 (2005)
72  et-0/0/3 Up 174142714 (0) 630008 (0)
73  et-0/0/5 Up 1209975390 (1) 398595 (1)
74  et-0/0/7 Up 3173525 (1003) 1210993710 (1003)
75  et-0/0/9 Up 0 (0) 66133 (0)
76  et-0/0/11 Up 0 (0) 66134 (0)
77  et-0/0/13 Up 0 (0) 66138 (0)
78  et-0/0/15 Up 0 (0) 66149 (0)

```

PIM-GW:

```

1  R6-PIM-GW> show pim join extensive
2  Instance: PIM.master Family: INET
3  R = Rendezvous Point Tree, S = Sparse, W = Wildcard
4
5  Group: 225.1.1.10
6  Source: *
7  RP: 1.1.1.7
8  Flags: sparse,rptree,wildcard
9  Upstream interface: Local
10 Upstream neighbor: Local
11 Upstream state: Local RP
12 Uptime: 1w4d 09:31:58
13 Downstream neighbors:
14 Interface: irb.2000 (assert winner)
15 80.1.1.50 State: Join Flags: SRW Timeout: 184
16 Uptime: 03:02:25 Time since last Join: 00:00:25
17 80.1.1.60 State: Join Flags: SRW Timeout: 184
18 Uptime: 03:01:25 Time since last Join: 00:00:25
19 Assert Winner: 80.1.1.70 Metric: 0 Pref: 2147483648 Timeout: 25
20 Interface: ge-0/0/2.1001
21 25.1.2.1 State: Join Flags: SRW Timeout: Infinity
22 Uptime: 05:21:57 Time since last Join: 03:02:37
23 Number of downstream interfaces: 2
24 Number of downstream neighbors: 3
25 Group: 225.1.1.10
26 Source: 25.1.1.2
27 Flags: sparse,spt
28 Upstream interface: ge-0/0/2.1000
29 Upstream neighbor: Direct
30 Upstream state: Local Source, Local RP, No Prune to RP
31 Keepalive timeout: 320
32 Uptime: 00:14:26
33 Downstream neighbors:
34 Interface: irb.2000
35 80.1.1.50 State: Join Flags: S Timeout: 184
36 Uptime: 00:14:26 Time since last Join: 00:00:25
37 80.1.1.60 State: Join Flags: S Timeout: 184

```

```

38      Uptime: 00:14:26 Time since last Join: 00:00:25
39      Interface: ge-0/0/2.1001
40      25.1.2.1 State: Join Flags: S Timeout: Infinity
41      Uptime: 00:14:26 Time since last Join: 00:14:26
42      Number of downstream interfaces: 2
43      Number of downstream neighbors: 3
44      <Output truncated for brevity>
45
46 R6-PIM-GW> show multicast route extensive
47 Instance: master Family: INET
48 Group: 225.1.1.10
49   Source: 25.1.1.2/32
50   Upstream interface: ge-0/0/2.1000
51   Downstream interface list:
52     ge-0/0/2.1001 irb.2000
53   Number of outgoing interfaces: 2
54   Session description: Unknown
55   Statistics: 491 kBps, 1002 pps, 909383 packets
56   Next-hop ID: 1048577
57   Upstream protocol: PIM
58   Route state: Active
59   Forwarding state: Forwarding
60   Cache lifetime/timeout: 360 seconds
61   Wrong incoming interface notifications: 0
62   Uptime: 00:15:18
63
64 Instance: master Family: INET6
65
66 R6-PIM-GW> monitor interface traffic
67 R6-re0 Seconds: 10 Time: 09:58:33
68
69 Interface Link Input packets (pps) Output packets (pps)
70 ge-0/0/0 Up 1116872970 (0) 3354149 (1001) >>>>>> Interface connected to BL1 (R4)
71 lc-0/0/0 Up 0 0
72 pfh-0/0/0 Up 0 0
73 ge-0/0/1 Up 305701 (0) 1209892515 (0)
74 ge-0/0/2 Up 1228962686 (1001) 1737588569 (1001) >>>>>> Interface connected to IXIA

```

Transmit State	Traffic Item Name	Enabled	Flow Groups	Tx Ports	Rx Ports	Endpoint/Encapsulation Sets
1	Source-R6-VLAN-1000	☒	1	1	3	1
2	Source-R0_VL_100	☐	1	1	4	1
3	Source-R1-VLAN-101	☐	1	1	3	1
4	Source-R6-VLAN-1001	☐	1	1	3	1

Rx Port	Tx Frames	Rx Expected Frames	Rx Frames	Frames Delta	Loss %	Tx Frame Rate	Rx Frame Rate	Tx L1 Rate (bps)	Rx L1 Rate (bps)	Rx Bytes	Tx Rate (Bps)	Rx Rate (Bps)	Tx Rate (bps)
1 R0-et-0/0/3 <-> R9 (Vlan2)	3,963	3,963	3,958	5	0.126	1,001.491	1,001.991	4,262,345.578	4,264,472.497	2,026,496	512,763.378	513,019.248	4,102,107.
2 R1-et-0/0/5 <-> R11 (Vlan4)	3,963	3,963	3,957	6	0.151	1,001.705	1,001.705	4,263,255.619	4,263,255.619	2,025,984	512,872.856	512,872.856	4,102,982.
3 R7-ge-0/0/1 <-> R10 (Vlan3)	3,963	3,963	3,957	6	0.151	1,001.846	1,001.846	4,263,854.556	4,263,854.556	2,025,984	512,944.909	512,944.909	4,103,559.

1.6 Configuration Snippets

1.6.1 Access / Underlay Configuration

SL1:

```

1 R0-SL1> show configuration interfaces
2 et-0/0/1 {
3   description "To Multicast Source/Receiver host"
4   flexible-vlan-tagging;
5   encapsulation flexible-ethernet-services;
6   unit 100 {
7     encapsulation vlan-bridge;
8     vlan-id 100;
9   }
10  unit 101 {
11    encapsulation vlan-bridge;
12    vlan-id 101;
13  }
14 }
15 et-0/0/3 {

```

```

16  description "To Multicast Source/Receiver host"
17  flexible-vlan-tagging;
18  encapsulation flexible-ethernet-services;
19  unit 100 {
20      encapsulation vlan-bridge;
21      vlan-id 100;
22  }
23  unit 101 {
24      encapsulation vlan-bridge;
25      vlan-id 101;
26  }
27  }
28  et-0/0/5 {
29      ether-options {
30          802.3ad ae1;
31      }
32  }
33  ae1 {
34      description "To CE-R7"
35      flexible-vlan-tagging;
36      encapsulation flexible-ethernet-services;
37      esi {
38          00:11:11:11:11:11:11:11:11:11;
39          all-active;
40      }
41      aggregated-ether-options {
42          lacp {
43              active;
44              system-id 00:11:11:11:11:11;
45          }
46      }
47      unit 100 {
48          encapsulation vlan-bridge;
49          vlan-id 100;
50      }
51      unit 101 {
52          encapsulation vlan-bridge;
53          vlan-id 101;
54      }
55  }
56  et-0/0/7 {
57      description "To Spine Leaf -- Overlay"
58      unit 0 {
59          family inet {
60              address 10.1.1.2/24;
61          }
62      }
63  }
64  et-0/0/9 {
65      description "To Spine Leaf -- Overlay"
66      unit 0 {
67          family inet {
68              address 11.1.1.2/24;
69          }
70      }
71  }
72  irb {
73      unit 100 {
74          virtual-gateway-accept-data;
75          family inet {
76              address 100.1.1.101/24 {
77                  preferred;
78                  virtual-gateway-address 100.1.1.254;
79              }
80          }
81      }
82      unit 101 {
83          virtual-gateway-accept-data;
84          family inet {
85              address 101.1.1.101/24 {
86                  preferred;
87                  virtual-gateway-address 101.1.1.254;
88              }
89          }
90      }
91      unit 900 {
92          virtual-gateway-accept-data;
93          family inet {
94              address 90.1.1.10/24 {
95                  virtual-gateway-address 90.1.1.254;
96              }

```

```

97     }
98   }
99 }
100 lo0 {
101   unit 0 {
102     family inet {
103       address 1.1.1.1/32;
104     }
105   }
106   unit 1 {
107     family inet {
108       address 1.1.10.1/32;
109     }
110   }

```

SL2:

```

1  R1-SL2> show configuration interfaces
2  et-0/0/5 {
3    description "To Multicast Source/Receiver host"
4    flexible-vlan-tagging;
5    encapsulation flexible-ethernet-services;
6    unit 100 {
7      encapsulation vlan-bridge;
8      vlan-id 100;
9    }
10   unit 101 {
11     encapsulation vlan-bridge;
12     vlan-id 101;
13   }
14 }
15 et-0/0/7 {
16   unit 0 {
17     family inet {
18       address 13.1.1.2/24;
19     }
20   }
21 }
22 et-0/0/3 {
23   ether-options {
24     802.3ad ael;
25   }
26 }
27 ael {
28   description "To CE-R7"
29   flexible-vlan-tagging;
30   encapsulation flexible-ethernet-services;
31   esi {
32     00:11:11:11:11:11:11:11:11:11;
33     all-active;
34   }
35   aggregated-ether-options {
36     lacp {
37       active;
38       system-id 00:11:11:11:11:11;
39     }
40   }
41   unit 100 {
42     encapsulation vlan-bridge;
43     vlan-id 100;
44   }
45   unit 101 {
46     encapsulation vlan-bridge;
47     vlan-id 101;
48   }
49 }
50 et-0/0/1 {
51   description "To Spine Leaf -- Overlay"
52   unit 0 {
53     family inet {
54       address 12.1.1.2/24;
55     }
56   }
57 }
58 irb {
59   unit 100 {
60     virtual-gateway-accept-data;
61     family inet {
62       address 100.1.1.102/24 {
63         virtual-gateway-address 100.1.1.254;

```

```

64     }
65   }
66 }
67 unit 101 {
68   virtual-gateway-accept-data;
69   family inet {
70     address 101.1.1.102/24 {
71       virtual-gateway-address 101.1.1.254;
72     }
73   }
74 }
75 unit 900 {
76   virtual-gateway-accept-data;
77   family inet {
78     address 90.1.1.20/24 {
79       virtual-gateway-address 90.1.1.254;
80     }
81   }
82 }
83 }
84 lo0 {
85   unit 0 {
86     family inet {
87       address 1.1.1.2/32;
88     }
89   }
90   unit 1 {
91     family inet {
92       address 1.1.10.2/32;
93     }
94   }
95 }

```

LS1 (R2):

```

1 regress@R2-re0-re0> show configuration
2 interfaces {
3   et-0/0/1 {
4     unit 0 {
5       family inet {
6         address 16.1.1.2/24;
7       }
8     }
9   }
10  et-0/0/3 {
11    unit 0 {
12      family inet {
13        address 15.1.1.2/24;
14      }
15    }
16  }
17  et-0/0/5 {
18    unit 0 {
19      family inet {
20        address 17.1.1.2/24;
21      }
22    }
23  }
24  et-0/0/7 {
25    unit 0 {
26      family inet {
27        address 12.1.1.1/24;
28      }
29    }
30  }
31  et-0/0/9 {
32    unit 0 {
33      family inet {
34        address 10.1.1.1/24;
35      }
36    }
37  }
38  lo0 {
39    unit 0 {
40      family inet {
41        address 1.1.1.3/32;
42      }
43    }
44  }
45 }

```

```

46 routing-options {
47     router-id 1.1.1.3;
48     autonomous-system 64512;
49 }
50 protocols {
51     ospf {
52         area 0.0.0.0 {
53             interface all;
54             interface re0:mgmt-0.0 {
55                 disable;
56             }
57         }
58     }
59 }

```

LS2 (R3):

```

1 regress@r2-re0# show interfaces
2
3 interfaces {
4     et-0/0/1 {
5         unit 0 {
6             family inet {
7                 address 16.1.1.1/24;
8             }
9         }
10    }
11    et-0/0/3 {
12        unit 0 {
13            family inet {
14                address 18.1.1.2/24;
15            }
16        }
17    }
18    et-0/0/5 {
19        unit 0 {
20            family inet {
21                address 14.1.1.2/24;
22            }
23        }
24    }
25    et-0/0/7 {
26        unit 0 {
27            family inet {
28                address 11.1.1.1/24;
29            }
30        }
31    }
32    et-0/0/9 {
33        unit 0 {
34            family inet {
35                address 13.1.1.1/24;
36            }
37        }
38    }
39    lo0 {
40        unit 0 {
41            family inet {
42                address 1.1.1.4/32;
43            }
44        }
45    }
46 }
47 routing-options {
48     router-id 1.1.1.4;
49     autonomous-system 64512;
50 }
51 protocols {
52     ospf {
53         area 0.0.0.0 {
54             interface all;
55             interface re0:mgmt-0.0 {
56                 disable;
57             }
58         }
59     }
60 }

```

BL1 (R4):

```

1 R4-BL1> show configuration interfaces
2 et-0/0/1 {
3     unit 0 {
4         family inet {
5             address 15.1.1.1/24;
6         }
7     }
8 }
9 et-0/0/3 {
10     unit 0 {
11         family inet {
12             address 14.1.1.1/24;
13         }
14     }
15 }
16 et-0/0/5 {
17     unit 0 {
18         family inet {
19             address 21.1.1.2/24;
20         }
21     }
22 }
23 et-0/0/7 {
24     unit 0 {
25         family inet {
26             address 19.1.1.2/24;
27         }
28     }
29 }
30 irb {
31     unit 100 {
32         virtual-gateway-accept-data;
33         family inet {
34             address 100.1.1.105/24 {
35                 virtual-gateway-address 100.1.1.254;
36             }
37         }
38     }
39     unit 101 {
40         virtual-gateway-accept-data;
41         family inet {
42             address 101.1.1.105/24 {
43                 virtual-gateway-address 101.1.1.254;
44             }
45         }
46     }
47     unit 900 {
48         virtual-gateway-accept-data;
49         family inet {
50             address 90.1.1.50/24 {
51                 virtual-gateway-address 90.1.1.254;
52             }
53         }
54     }
55 }
56 lo0 {
57     unit 0 {
58         family inet {
59             address 1.1.1.5/32;
60         }
61     }
62     unit 1 {
63         family inet {
64             address 1.1.50.2/32;
65         }
66     }
67 }

```

BL2 (R5):

```

1 R5-BL2> show configuration interfaces
2 et-0/0/1 {
3     unit 0 {
4         family inet {
5             address 18.1.1.1/24;
6         }
7     }
8 }
9 et-0/0/3 {

```

```

10     unit 0 {
11         family inet {
12             address 17.1.1.1/24;
13         }
14     }
15 }
16 et-0/0/5 {
17     unit 0 {
18         family inet {
19             address 22.1.1.2/24;
20         }
21     }
22 }
23 et-0/0/7 {
24     unit 0 {
25         family inet {
26             address 19.1.1.1/24;
27         }
28     }
29 }
30 irb {
31     unit 100 {
32         virtual-gateway-accept-data;
33         family inet {
34             address 100.1.1.106/24 {
35                 virtual-gateway-address 100.1.1.254;
36             }
37         }
38     }
39     unit 101 {
40         virtual-gateway-accept-data;
41         family inet {
42             address 101.1.1.106/24 {
43                 virtual-gateway-address 101.1.1.254;
44             }
45         }
46     }
47     unit 900 {
48         virtual-gateway-accept-data;
49         family inet {
50             address 90.1.1.60/24 {
51                 virtual-gateway-address 90.1.1.254;
52             }
53         }
54     }
55 }
56 lo0 {
57     unit 0 {
58         family inet {
59             address 1.1.1.6/32;
60         }
61     }
62     unit 1 {
63         family inet {
64             address 1.1.60.2/32;
65         }
66     }
67 }

```

1.6.2 Overlay Configuration

SL1 (R0):

```

1 R0-SL1> show configuration protocols
2 bgp {
3     group ibgp {
4         local-address 1.1.1.1;
5         family evpn {
6             signaling;
7         }
8         peer-as 64512;
9         neighbor 1.1.1.2;
10        neighbor 1.1.1.5;
11        neighbor 1.1.1.6;
12    }
13 }

```

```

14 ospf {
15     area 0.0.0.0 {
16         interface all;
17         interface re0:mgmt-0.0 {
18             disable;
19         }
20     }
21 }
22 lldp {
23     interface all;
24 }
25
26 R0-SL1# show routing-options
27 router-id 1.1.1.1;
28 autonomous-system 64512;
29 forwarding-table {
30     chained-composite-next-hop {
31         ingress {
32             evpn;
33         }
34     }
35 }

```

SL2 (R1):

```

1 R1-SL2> show configuration protocols
2 bgp {
3     group ibgp {
4         local-address 1.1.1.2;
5         family evpn {
6             signaling;
7         }
8         peer-as 64512;
9         neighbor 1.1.1.1;
10        neighbor 1.1.1.5;
11        neighbor 1.1.1.6;
12    }
13 }
14 ospf {
15     area 0.0.0.0 {
16         interface all;
17         interface re0:mgmt-0.0 {
18             disable;
19         }
20     }
21 }
22 lldp {
23     interface all;
24 }
25 R1-SL2# show routing-options
26 router-id 1.1.1.2;
27 autonomous-system 64512;
28 forwarding-table {
29     chained-composite-next-hop {
30         ingress {
31             evpn;
32         }
33     }
34 }

```

BL1:

```

1 R4-BL1> show configuration protocols
2 bgp {
3     group ibgp {
4         local-address 1.1.1.5;
5         family evpn {
6             signaling;
7         }
8         peer-as 64512;
9         neighbor 1.1.1.1;
10        neighbor 1.1.1.2;
11        neighbor 1.1.1.6;
12    }
13 }
14 ospf {
15     area 0.0.0.0 {
16         interface all;
17         interface re0:mgmt-0.0 {

```

```

18         disable;
19     }
20 }
21 }
22 R4-BL1> show configuration routing-options
23 router-id 1.1.1.5;
24 autonomous-system 64512;
25 forwarding-table {
26     chained-composite-next-hop {
27         ingress {
28             evpn;
29         }
30     }
31 }

```

BL2:

```

1 R5-BL2> show configuration protocols
2 bgp {
3     group ibgp {
4         local-address 1.1.1.6;
5         family evpn {
6             signaling;
7         }
8         peer-as 64512;
9         neighbor 1.1.1.1;
10        neighbor 1.1.1.2;
11        neighbor 1.1.1.5;
12    }
13 }
14 ospf {
15     area 0.0.0.0 {
16         interface all;
17         interface re0:mgmt-0.0 {
18             disable;
19         }
20     }
21 }
22 R5-BL2> show configuration routing-options
23 router-id 1.1.1.6;
24 autonomous-system 64512;
25 forwarding-table {
26     chained-composite-next-hop {
27         ingress {
28             evpn;
29         }
30     }
31 }

```

1.6.3 CE Configuration - SP/EP style

CE1 – SP Style:

```

1 R7-re0> show configuration
2 interfaces {
3     ge-0/0/0 {
4         gigether-options {
5             802.3ad ael;
6         }
7     }
8     ge-0/0/1 {
9         flexible-vlan-tagging;
10        encapsulation flexible-ethernet-services;
11        unit 100 {
12            encapsulation vlan-bridge;
13            vlan-id 100;
14        }
15        unit 101 {
16            encapsulation vlan-bridge;
17            vlan-id 101;
18        }
19    }
20    ge-0/0/2 {
21        gigether-options {
22            802.3ad ael;

```

```

23     }
24 }
25 ael {
26     flexible-vlan-tagging;
27     encapsulation flexible-ethernet-services;
28     aggregated-ether-options {
29         lacp {
30             active;
31             system-id 00:11:11:11:11:12;
32         }
33     }
34     unit 100 {
35         encapsulation vlan-bridge;
36         vlan-id 100;
37     }
38     unit 101 {
39         encapsulation vlan-bridge;
40         vlan-id 101;
41     }
42 }
43 }
44 bridge-domains {
45     bd-100 {
46         vlan-id 100;
47         interface ge-0/0/1.100;
48         interface ael.100;
49     }
50     bd-101 {
51         vlan-id 101;
52         interface ge-0/0/1.101;
53         interface ael.101;
54     }
55 }

```

CE1 – EP Style:

```

1 R7-re0> show configuration
2 interfaces {
3     ge-0/0/0 {
4         gigerther-options {
5             802.3ad ael;
6         }
7     }
8     ge-0/0/1 {
9         unit 0 {
10             family ethernet-switching;
11             vlan members [v100 v101];
12         }
13     }
14     ge-0/0/2 {
15         gigerther-options {
16             802.3ad ael;
17         }
18     }
19     ael {
20         aggregated-ether-options {
21             lacp {
22                 active;
23                 system-id 00:11:11:11:11:12;
24             }
25         }
26         unit 0 {
27             family ethernet-switching;
28             vlan members [v100 v101];
29         }
30     }
31 }
32 bridge-domains {
33     v100 {
34         vlan-id 100;
35         interface ge-0/0/1.100;
36         interface ael.100;
37     }
38     v101 {
39         vlan-id 101;
40         interface ge-0/0/1.101;
41         interface ael.101;
42     }
43 }

```

1.7 Glossary

- AR: Assisted Replication
- BDF: Backup Designated Forwarder
- BGP: Border Gateway Protocol
- BL: Border Leal
- CE: Customer Edge
- CE Mesh: CE side mesh group
- DF: Designated Forwarder
- ESI: Ethernet Segment Identifier
- EVPN: Ethernet Virtual Private Network
- EVI: EVPN Instance
- IRB: Integrated Routing and Bridging
- LACP: Link Aggregation Control Protocol
- LAG: Link Aggregation
- MCSNOOPD: Multicast Snooping Daemon responsible for handling the IGMP Snooping
- OIF: Outgoing Interface
- OISM: Optimized Inter Subnet multicast
- PE: Provider Edge
- PEG: PIM EVPN Gateway
- SBD: Supplement Bridge domain
- RR: Route Reflector
- SL: Server Leaf
- SMET: Selective Multicast Ethernet
- VPN: Virtual Private Network
- VE Mesh: EVPN Core side mesh group

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